

MultiCaRe Taxonomy Documentation

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Ontology Overview

Name: MultiCaRe Taxonomy

Acronym: MCR_TX

Creator: Mauro Nievas Offidani, MD, MSc

Description: This document provides details about the MultiCaRe Taxonomy, the class structure used for medical image classification in the MultiCaRe dataset. It includes illustrations of partial class maps, class descriptions, and image examples per class.

Domain: Medical Imaging.

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Ontology Development: The MultiCaRe Taxonomy was developed using captions from over 150,000 images in the MultiCaRe dataset, sourced from case reports in PubMed Central. It focuses on categories related to image types, anatomical regions, and views, with only a few classes related to image findings. Classes absent or extremely rare in image captions were excluded.

This taxonomy adheres to the principles of Multiplex Classification, an approach designed for problems involving any number of classes and logical constraints. The class structure is a Disjoint-Union-Based Tree (DUBT), where each class (except the root) belongs to a disjoint union set, and parent classes may represent unions of multiple disjoint subsets. The is-a relation is the only relation type present in this taxonomy.

Use Cases: This taxonomy can be used to classify medical images according to their type (radiology, pathology, ct, ultrasound, etc), their attributes (doppler, contrast, etc) and the corresponding anatomical region (such as thorax or head) and view (axial, frontal, etc).

Class annotations:

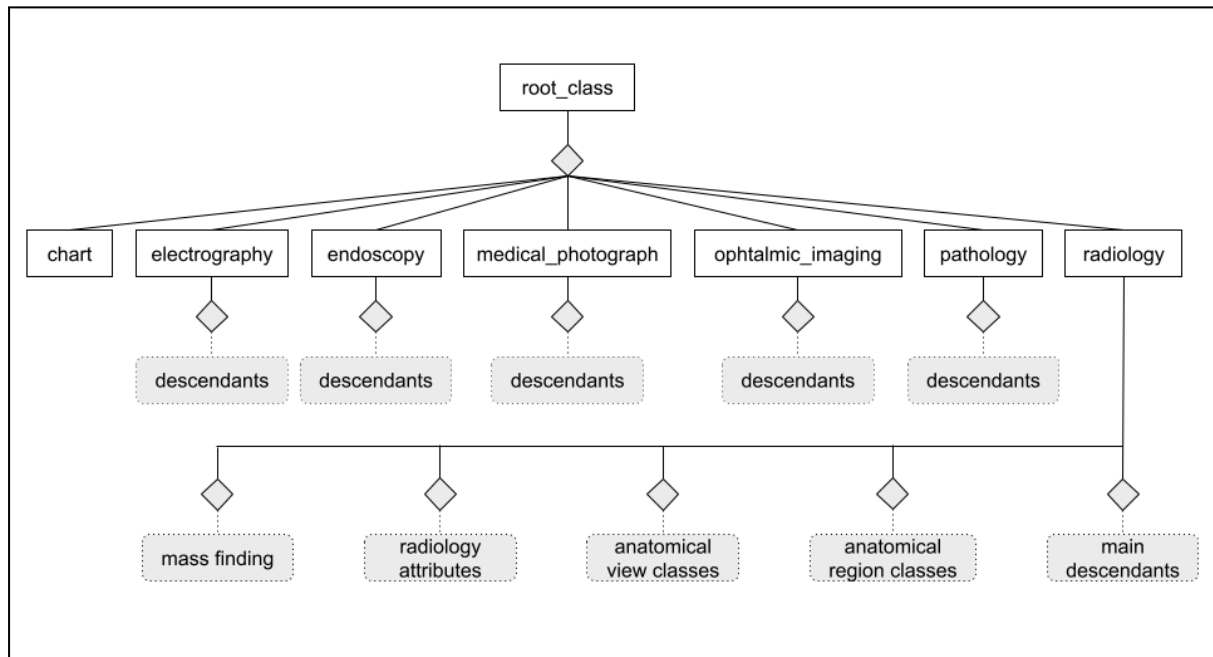
- **description:** Definition of the class.
- **synonyms:** Terms that mean the same as the given class.
- **SNOMED_ID:** Link to the equivalent class from SNOMED CT, for interoperability purposes.
- **auxiliary_class:** 'True' means that the class was created to meet the requirements of the Multiplex Classification Framework (this is the case of 'no_doppler' and other negative classes from binary classification tasks).
- **class_path:** String with a unique identifier used in the original file, which includes the information about all the ascendent classes from the given class together with their corresponding trees. ':' is used to separate a tree and its classes, '.' is used to separate a class from its subclass, and '~' is used to separate a class and its subsidiary tree.
- **associated_compound_classes:** List of compound classes associated with the given class ('_none' is the default value if there is no associated class). An instance belongs to a given associated class if it belongs to all the classes that have such

class under this data field. For example, if the classes 'heart' and 'ultrasound' have the associated class 'echocardiogram', then any instance that belongs both to 'heart' and 'ultrasound' should be classified also as 'echocardiogram'.

- **preprocessed_from:** List of all the classes present in the initial taxonomy that were merged into the given class during preprocessing ('_none' by default). For example, if the class 'ct_scan' has 'ct' and 'computed_tomography' as 'preprocessed_from', it means that the original taxonomy included such classes, and their names were merged into 'ct_scan' (which may also have been present originally).
- **tree_name:** This taxonomy was created from a file including different class trees. This field refers to the tree_name present in that source file.

root_class descendants

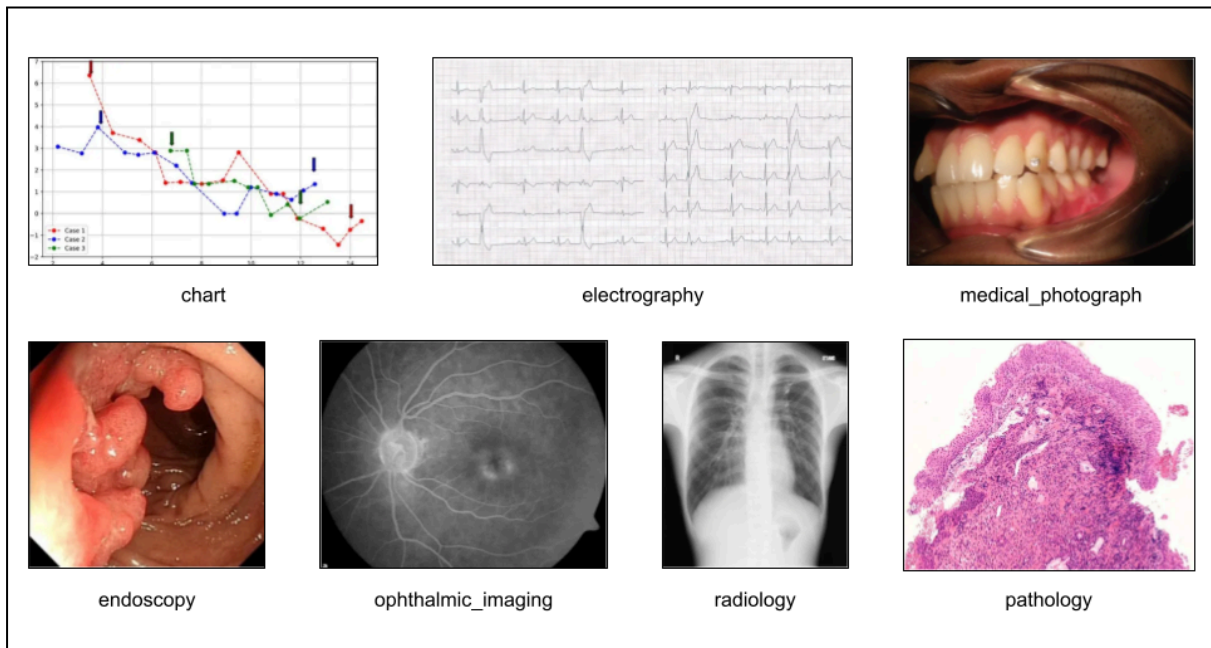
Partial class map



Class descriptions

- **chart:** Visual representations of data, often used to summarize patient information, trends, or results. Examples include graphs, tables, and other diagrammatic data displays.
- **electrography:** Imaging or recordings based on electrical activity in the body, such as electrocardiograms (ECG) or electroencephalograms (EEG).
- **endoscopy:** Internal imaging obtained using endoscopic devices, capturing detailed views of organs or tissues via minimally invasive techniques. Examples: colonoscopy, bronchoscopy.
- **medical_photograph:** Photographic images used for documentation of clinical conditions or procedures.
- **ophthalmic_imaging:** Images specifically focused on the eye and surrounding structures, typically captured using modalities such as fundus photography, optical coherence tomography (OCT), or slit-lamp imaging.
- **pathology:** Images of tissues and cells, often obtained using microscopes. Examples include histopathological slides and cytological preparations.
- **radiology:** Medical images acquired using radiological techniques, including X-rays, computed tomography (CT), magnetic resonance imaging (MRI), ultrasound, and nuclear medicine scans.

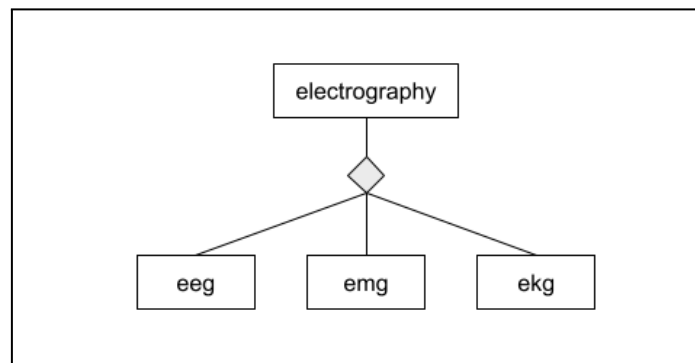
Image examples



Child classes of root_class.

electrography descendants

Partial class map



Class descriptions

- **eeg**: Electroencephalography (EEG) captures the brain's electrical activity using electrodes placed on the scalp. Commonly used for diagnosing neurological conditions such as epilepsy, sleep disorders, or brain injuries.
- **emg**: Electromyography (EMG) records the electrical activity of muscles during rest and movement. It is used to evaluate neuromuscular health and diagnose conditions like peripheral neuropathy or myopathy.
- **ekg**: Electrocardiography (EKG) measures the heart's electrical activity over time through electrodes on the skin. It is essential for diagnosing heart conditions like arrhythmias, myocardial infarction, and other cardiac abnormalities.

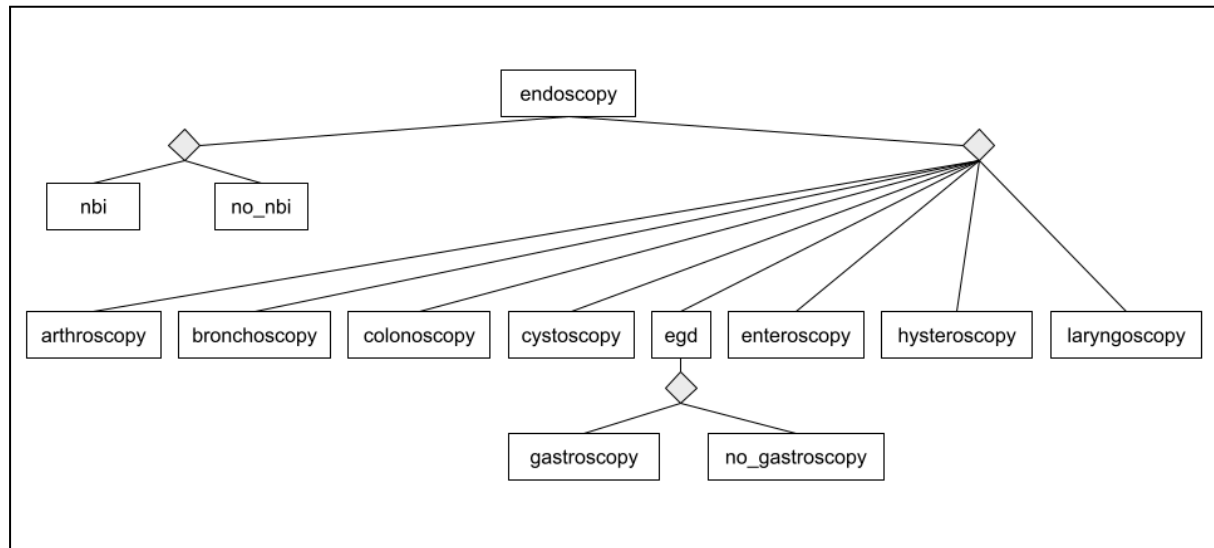
Image examples



Child classes of electrography.

endoscopy descendants

Partial class map

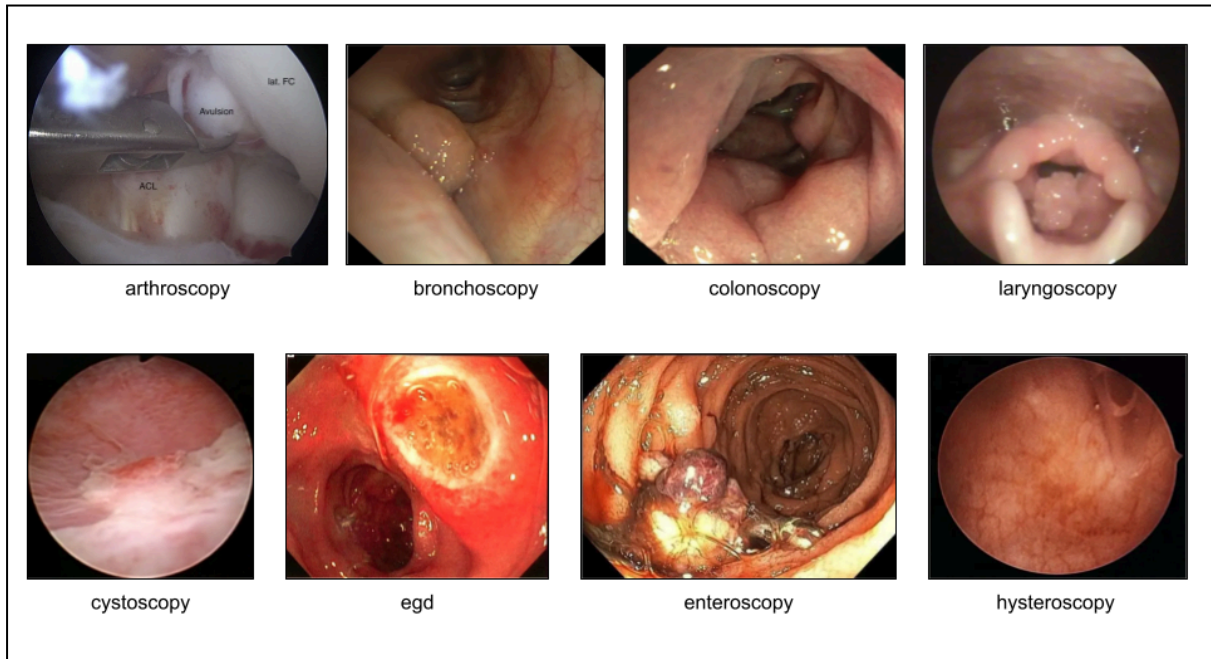


Class descriptions

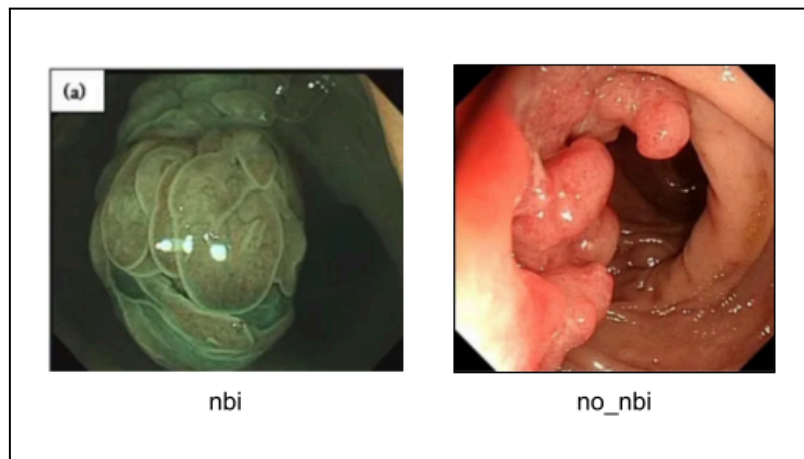
- **arthroscore:** A minimally invasive procedure for examining and treating joints using an arthroscope. Commonly used for knees, shoulders, and other joints to diagnose and repair injuries.
- **bronchoscore:** A procedure using a bronchoscope to examine the airways and lungs, often used to diagnose or treat conditions like tumors, infections, or obstructions.
- **colonoscore:** An endoscopic procedure for examining the colon and rectum. Typically used for screening and diagnosing conditions like colorectal cancer, polyps, or inflammatory bowel disease.
- **cystoscore:** An endoscopic procedure for examining the bladder and urethra, often used to diagnose urinary tract disorders, bladder cancer, or blockages.
- **egd:** Esophagogastroduodenoscopy (EGD) is an upper endoscopic procedure used to examine the esophagus, stomach, and duodenum for conditions like ulcers, GERD, or tumors.
- **gastroscopy:** A specific type of upper endoscopy focusing on the stomach, often used for diagnosing gastritis, ulcers, or tumors.
- **no_gastroscopy:** It refers to any EGD image that is not a gastroscopy (such as duodenoscopy).
- **enteroscore:** An endoscopic procedure for examining the small intestine, typically used for identifying causes of gastrointestinal bleeding or small bowel abnormalities.
- **hysteroscore:** A procedure using a hysteroscope to examine the inside of the uterus, often for diagnosing or treating conditions like fibroids, polyps, or abnormal bleeding.
- **laryngoscore:** A procedure to visualize the larynx and vocal cords, typically used to diagnose voice disorders, throat pain, or airway blockages.

- **nbi**: Narrow Band Imaging (NBI) is an advanced endoscopic imaging technique that enhances visualization of blood vessels and mucosal patterns, often used to detect early cancer.
- **no_nbi**: Negative class that refers to any endoscopy that does not belong to the nbi class.

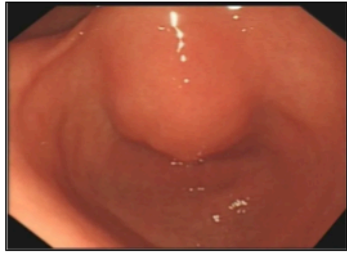
Image examples



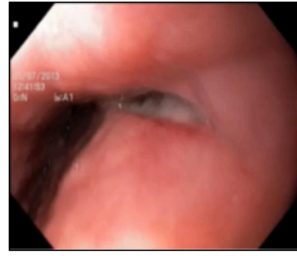
Child classes of endoscopy (main descendants).



Child classes of endoscopy (nbi attribute).



gastroscopy

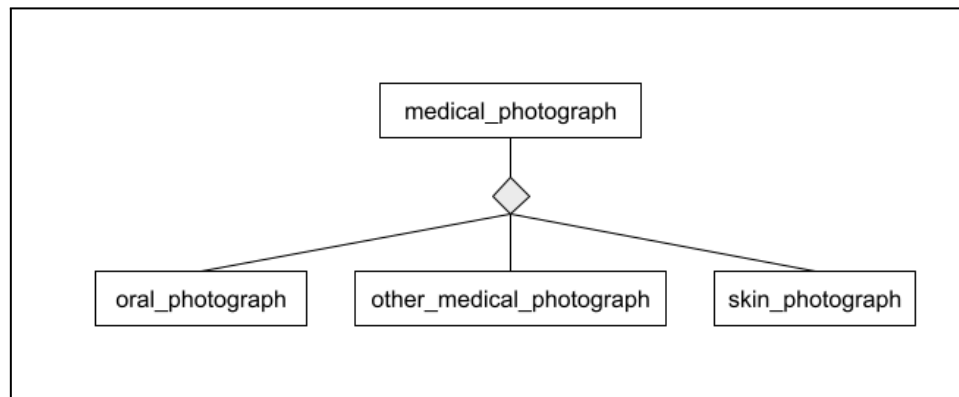


no_gastroscopy

Child classes of egd class.

medical_photograph descendants

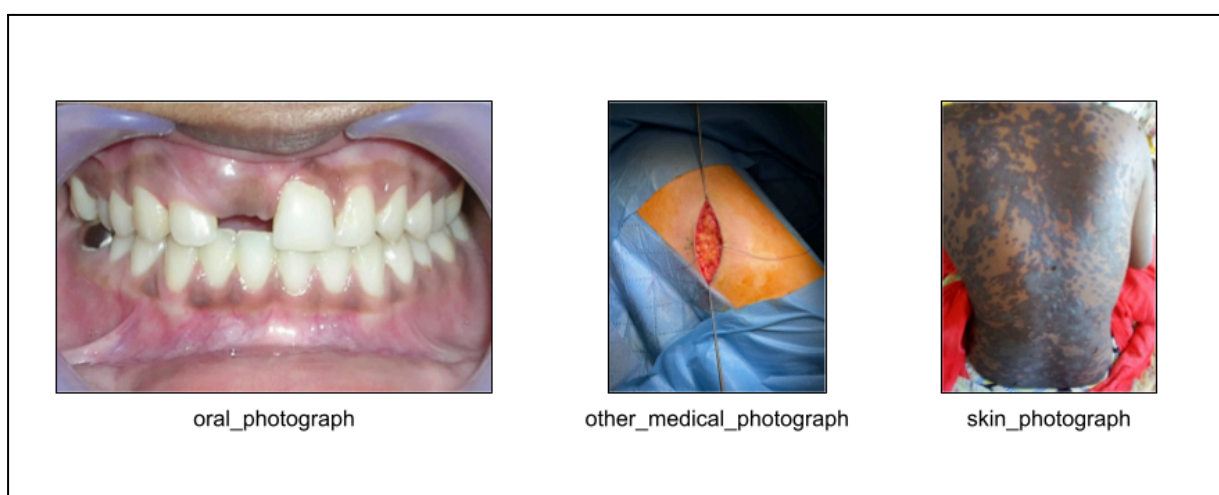
Partial class map



Class descriptions

- **oral_photograph:** Clinical photographs of the oral cavity, including teeth, gums, tongue, and other intraoral structures. Commonly used in dentistry and oral surgery for diagnosis, treatment planning, and documentation.
- **skin_photograph:** Clinical photographs of the skin, used for documenting dermatological conditions such as rashes, lesions, wounds, or skin infections. Commonly employed in dermatology and wound care.
- **other_medical_photograph:** Medical photographs that do not fall into specific categories like skin or oral photographs. These may include images of surgical sites, or medical devices.

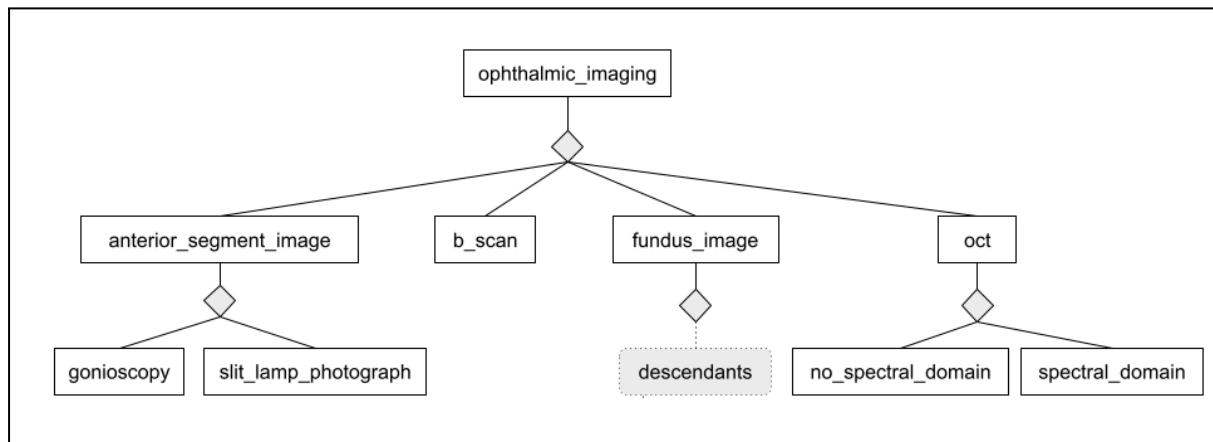
Image examples



Child classes of medical_photograph.

ophthalmic_imaging descendants

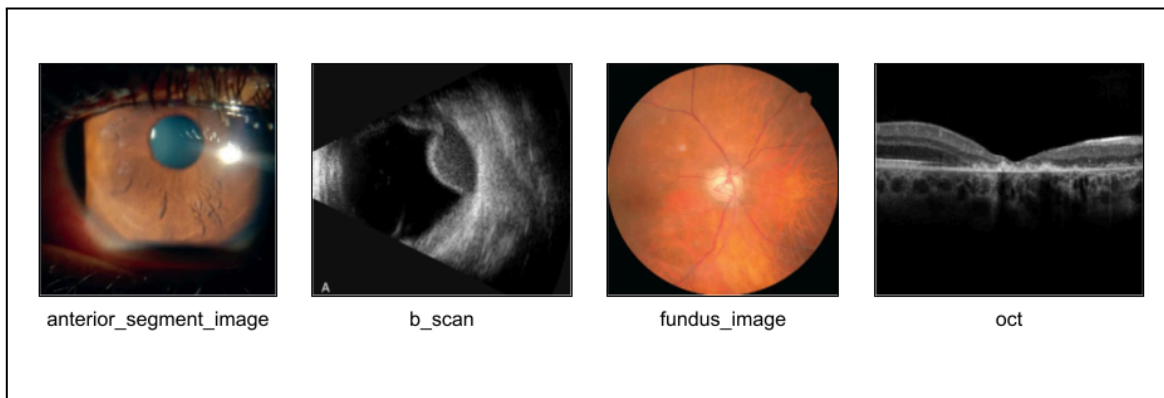
Partial class map



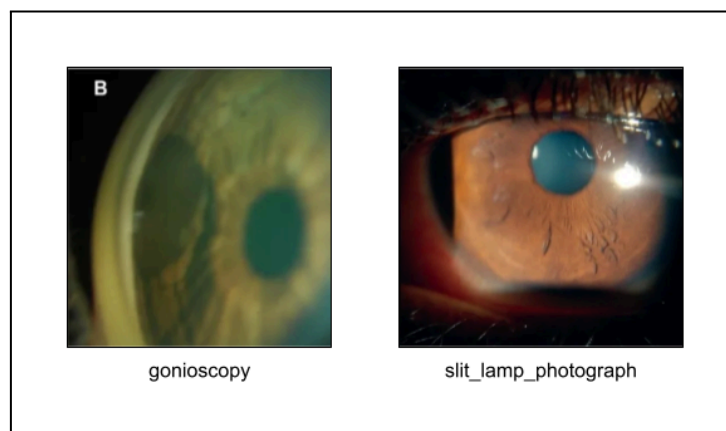
Class descriptions

- **anterior_segment_image:** Images of the anterior segment of the eye, including structures such as the cornea, iris, lens, and anterior chamber. Commonly used in ophthalmology to evaluate conditions like cataracts or corneal diseases.
- **gonioscopy:** Specialized imaging or visual assessment of the anterior chamber angle, used to evaluate and diagnose glaucoma and related conditions.
- **slit_lamp_photograph:** High-resolution photographs taken using a slit lamp microscope to document external and anterior segment structures of the eye, such as the cornea, conjunctiva, and iris.
- **b_scan:** B-Scan images are ultrasound images of the eye and orbit, often used when visualization is hindered by opaque media, such as cataracts or hemorrhage, to evaluate retinal or orbital conditions.
- **fundus_image:** Photographs of the interior surface of the eye, including the retina, optic disc, macula, and blood vessels, commonly used for diagnosing diabetic retinopathy, macular degeneration, and other retinal disorders.
- **oct:** Optical Coherence Tomography (OCT) is a non-invasive imaging technique that provides high-resolution cross-sectional images of retinal and other ocular structures. Widely used in the diagnosis and management of retinal and macular diseases.
- **spectral_domain:** Refers to Spectral Domain Optical Coherence Tomography (SD-OCT), an advanced type of OCT that provides detailed, high-speed imaging of retinal and other ocular structures.
- **no_spectral_domain:** Is a negative class for oct images that do not belong to the spectral_domain class.

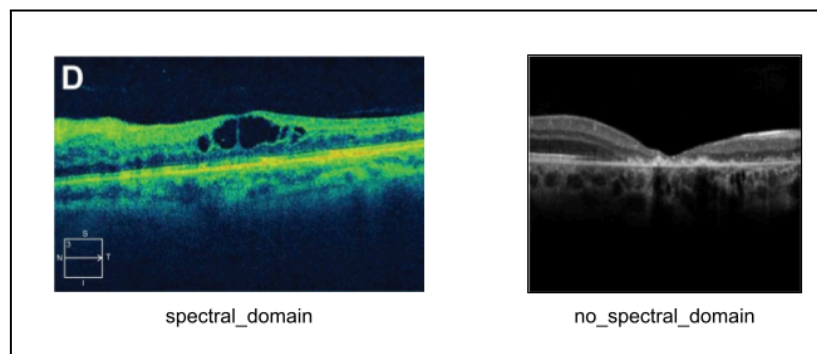
Image examples



Child classes of ophthalmic_imaging.



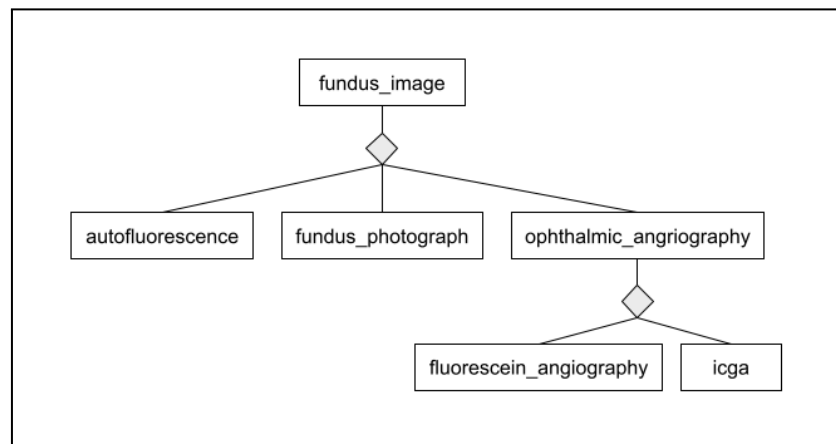
Child classes of anterior_segment_image.



Child classes of oct class.

fundus_image descendants

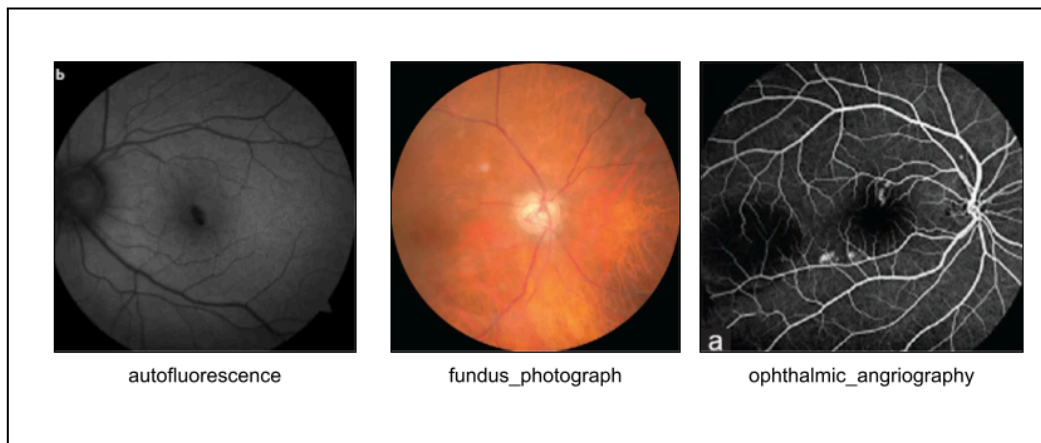
Partial class map



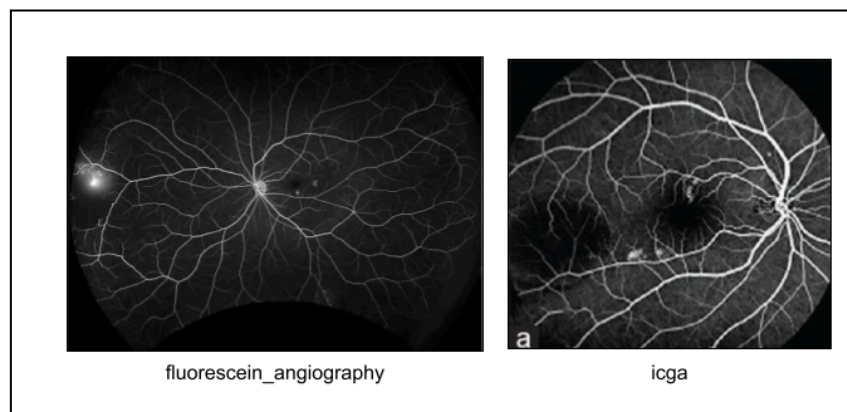
Class descriptions

- **autofluorescence:** A non-invasive imaging technique that captures natural fluorescence emitted by certain eye structures, primarily used to evaluate retinal and macular health, such as detecting lipofuscin in retinal pigment epithelium.
- **fundus_photograph:** Standard color photographs of the retina and optic disc, providing a detailed view of the posterior segment of the eye. Commonly used for diagnosing retinal diseases like diabetic retinopathy and glaucoma.
- **ophthalmic_angiography:** General term for imaging techniques that use dye-based methods to visualize blood flow in the retina and choroid, including fluorescein and indocyanine green angiography.
- **fluorescein_angiography:** A specific type of ophthalmic angiography that uses fluorescein dye to highlight retinal blood vessels, aiding in the diagnosis of conditions such as macular degeneration, diabetic retinopathy, and retinal vein occlusions.
- **icga:** Indocyanine Green Angiography (ICGA) is an angiographic technique that uses indocyanine green dye to visualize the choroidal vasculature. It is particularly useful for diagnosing conditions such as choroidal neovascularization and central serous chorioretinopathy.

Image examples



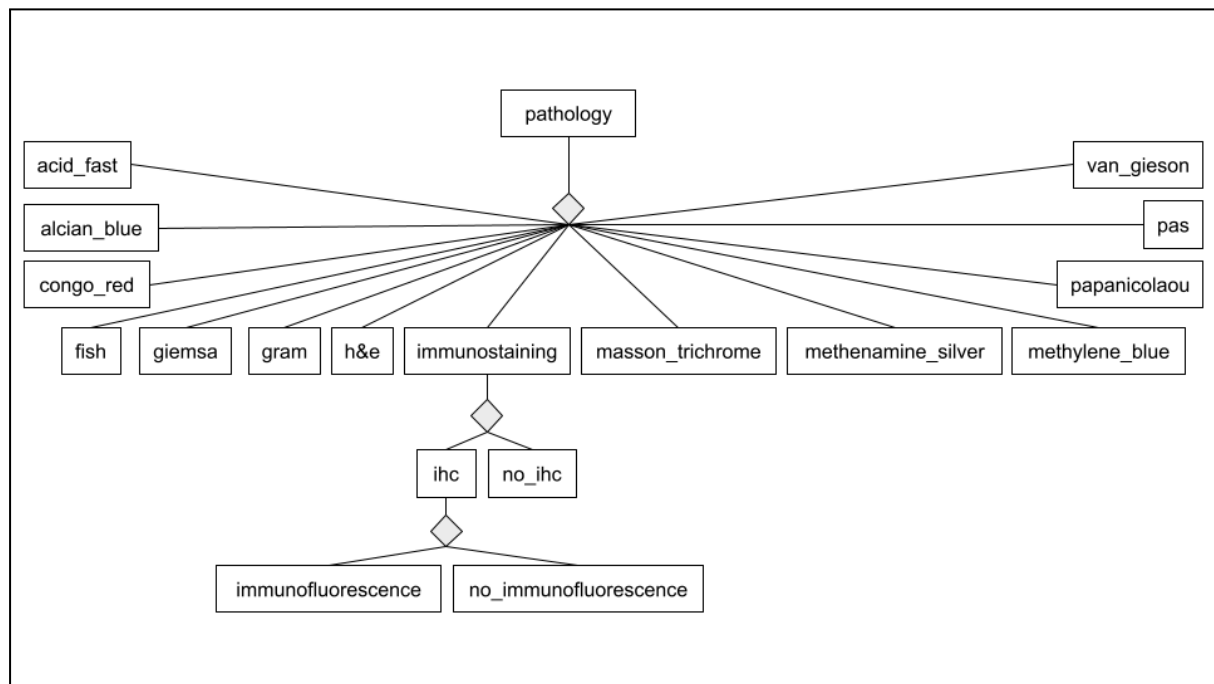
Child classes of fundus_image.



Child classes of ophthalmic_angiography.

pathology descendants

Partial class map

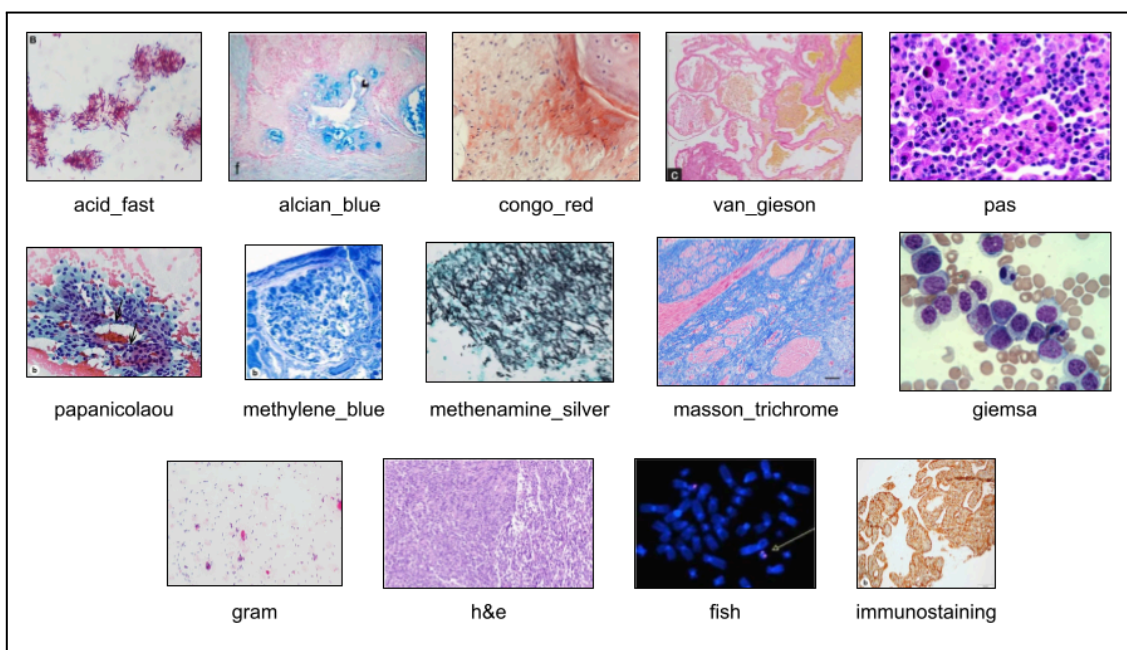


Class descriptions

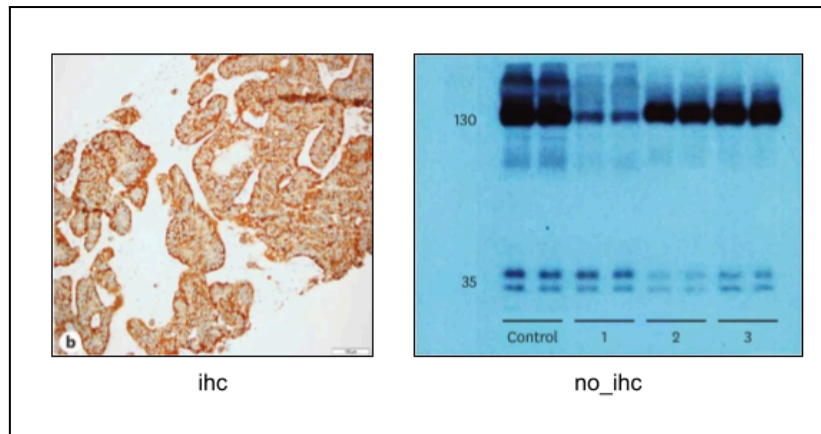
- **acid_fast:** A staining technique used to identify acid-fast bacteria, such as *Mycobacterium tuberculosis*. The method highlights bacteria that retain the dye even after exposure to acid-alcohol.
- **alcian_blue:** A histological stain used to detect acidic mucopolysaccharides and glycosaminoglycans, often employed for studying cartilage, connective tissues, and epithelial mucins.
- **congo_red:** A dye primarily used for detecting amyloid deposits in tissues. Under polarized light, amyloid stained with Congo Red exhibits apple-green birefringence.
- **fish:** Fluorescence In Situ Hybridization (FISH) is a molecular technique that uses fluorescent probes to identify specific DNA or RNA sequences in tissues or cells, aiding in genetic and chromosomal analysis.
- **giemsa:** A stain used in cytology and hematology for identifying blood cells, parasites (e.g., *Plasmodium*), and microorganisms. It is also used for differential blood smears and bone marrow evaluations.
- **gram:** A staining technique to classify bacteria into Gram-positive (purple) or Gram-negative (pink) based on cell wall structure. Essential for microbial diagnostics.
- **h&e:** Hematoxylin and Eosin (H&E) is the most commonly used histological stain, providing contrast between cell nuclei (blue/purple) and cytoplasm or extracellular matrix (pink).
- **immunostaining:** A general term for techniques that use antibodies to detect specific antigens in tissues or cells. Includes methods like immunohistochemistry and immunofluorescence.

- **ihc:** Immunohistochemistry (IHC) is a specific type of immunostaining that uses antibodies and chromogenic substrates to visualize antigen distribution in tissue sections.
- **no_ihc:** This is a negative class for images that belong to the immunostaining class but not to the ihc class.
- **immunofluorescence:** A technique that uses fluorescently labeled antibodies to detect specific antigens in tissues or cells. It is widely used in research and diagnostics for cellular and molecular analysis.
- **no_immunofluorescence:** This is a negative class for images that belong to the ihc class but not to the immunofluorescence class.
- **masson_trichrome:** A multi-color stain used to differentiate muscle, collagen, and fibrin in tissues. It highlights collagen in blue or green, cytoplasm in red, and nuclei in dark blue.
- **methamine_silver:** A staining method used to identify fungi and basement membranes. Commonly used in diagnosing infections like *Pneumocystis jirovecii* or visualizing glomerular structures.
- **methylene_blue:** A basic stain used for highlighting cell nuclei, bacterial cells, and some tissue structures. Commonly employed in bacteriology and cytology.
- **papanicolaou:** A cytological stain used in the Pap test for cervical cancer screening. It differentiates between basophilic, eosinophilic, and neutral cell components.
- **pas:** Periodic Acid-Schiff (PAS) stain is used to detect polysaccharides, glycogen, and mucosubstances in tissues. It is particularly useful for identifying fungal infections and basement membranes.
- **van_gieson:** A stain used to differentiate collagen fibers (red) from muscle and cytoplasm (yellow) in histological sections. Often used in connective tissue studies.

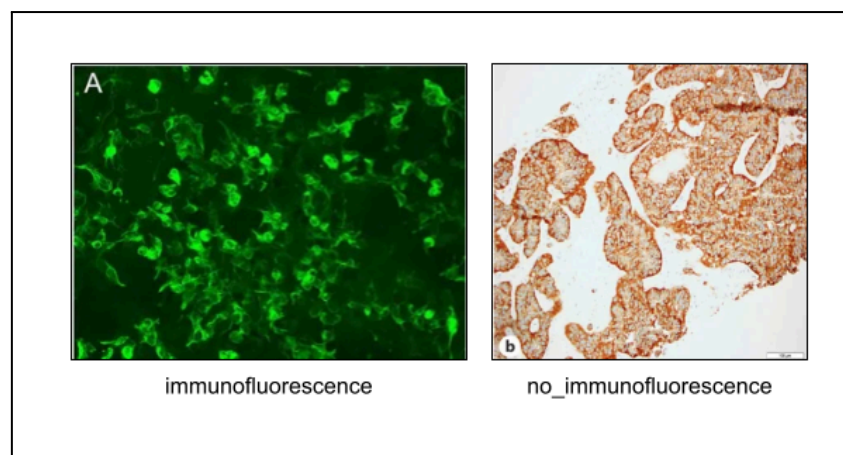
Image examples



Child classes of pathology.



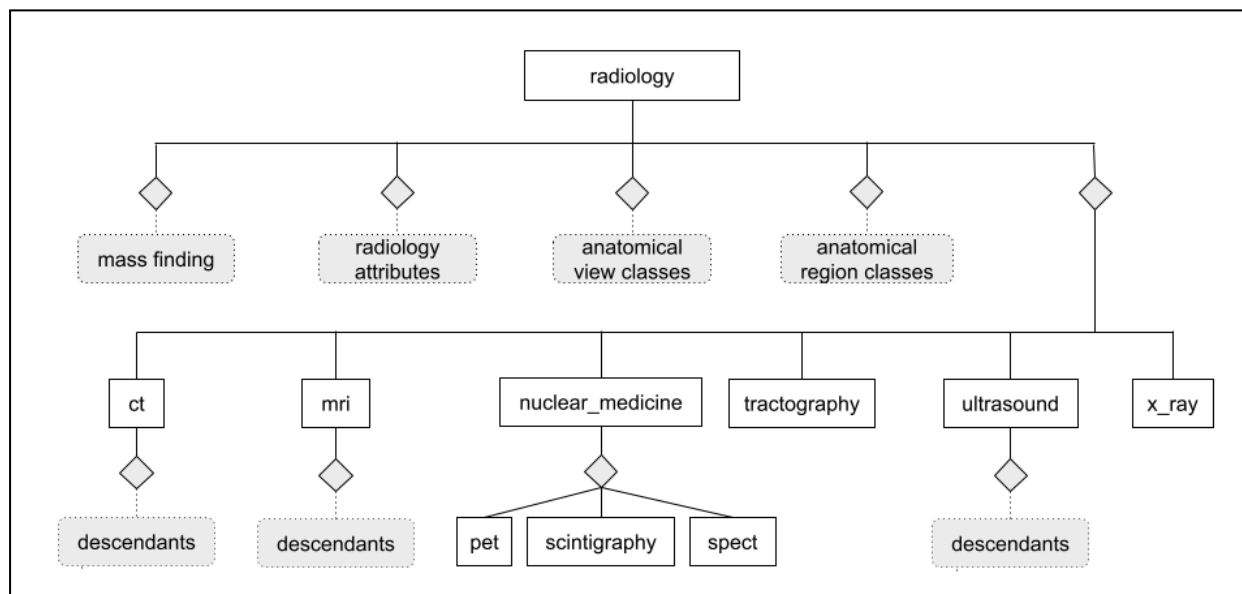
Child classes of immunostaining.



Child classes of ihc class.

radiology main descendants

Partial class map

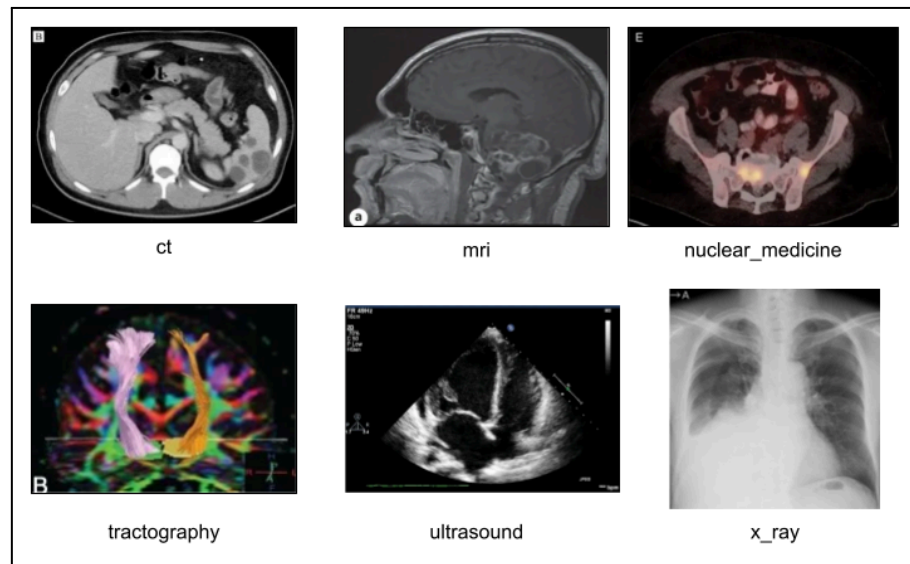


Class descriptions

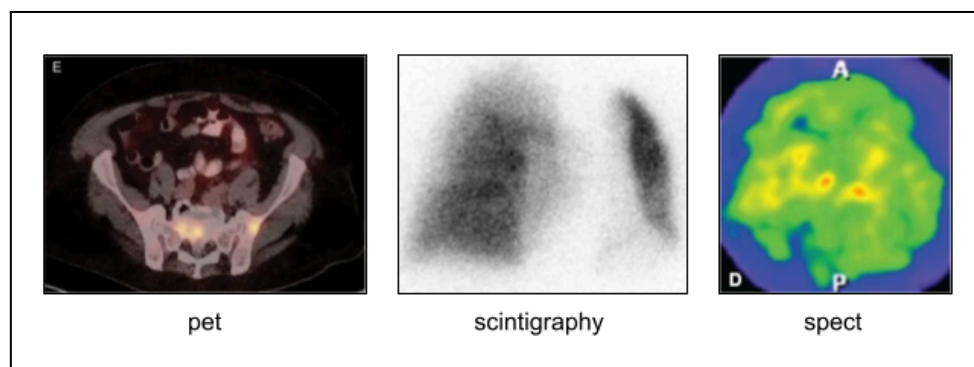
- **ct:** Computed Tomography (CT) is a medical imaging technique that uses X-rays to create cross-sectional images of the body. It is commonly used for detecting tumors, injuries, infections, and internal bleeding.
- **mri:** Magnetic Resonance Imaging (MRI) uses strong magnetic fields and radio waves to create detailed images of organs and tissues. It is particularly useful for imaging the brain, spine, muscles, and soft tissues.
- **nuclear_medicine:** A medical imaging technique that uses small amounts of radioactive material to diagnose and treat diseases. It provides functional imaging, unlike other modalities that focus on anatomical structures.
- **pet:** Positron Emission Tomography (PET) is a nuclear medicine imaging technique that detects metabolic activity in tissues. It is often used for cancer detection, brain disorders, and cardiac assessments.
- **scintigraphy:** A type of nuclear medicine imaging that uses a gamma camera to capture images based on the distribution of a radiopharmaceutical within the body. Commonly used to assess organ function and detect abnormalities.
- **spect:** Single Photon Emission Computed Tomography (SPECT) is a nuclear imaging technique that provides 3D images of the body by detecting gamma rays emitted by a radiotracer. It is used for brain, heart, and bone imaging.
- **tractography:** A specialized MRI technique used to visualize white matter tracts in the brain. It is commonly used in neurosurgery and research to map brain connectivity and identify important neural pathways.
- **ultrasound:** A non-invasive imaging technique that uses sound waves to create real-time images of internal organs and tissues. It is commonly used in obstetrics, cardiology, and musculoskeletal imaging.

- **x_ray**: A medical imaging technique that uses X-ray radiation to create images of the internal structures of the body, commonly used for bone fractures, lung conditions, and dental exams.

Image examples



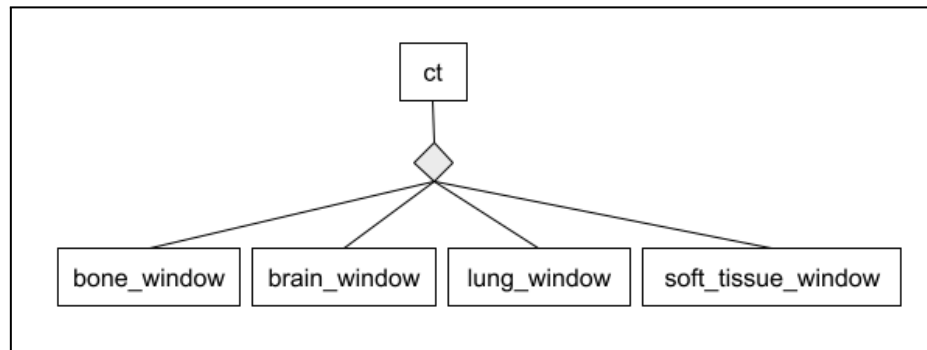
Child classes of radiology.



Child classes of nuclear_medicine.

ct descendants

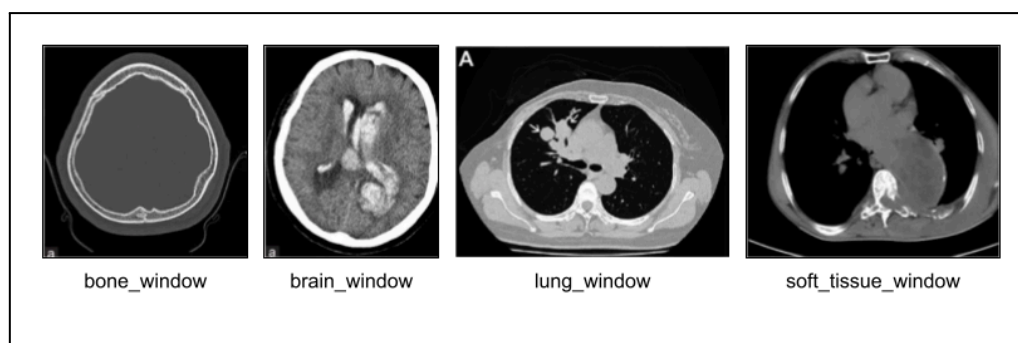
Partial class map



Class descriptions

- **bone_window:** A CT imaging window setting that enhances the contrast of bone structures, making them more visible while reducing the visibility of soft tissues. It is commonly used for evaluating fractures, bone lesions, and bone density.
- **brain_window:** A CT imaging window setting optimized to enhance the contrast of brain tissue structures, particularly for detecting abnormalities like tumors, strokes, and hemorrhages. It provides a clear view of the brain's internal architecture.
- **lung_window:** A CT imaging window setting that optimizes the visibility of lung tissues and air spaces, useful for detecting lung conditions such as pneumonia, tumors, or emphysema. It reduces the visibility of bones and soft tissues to highlight lung structures.
- **soft_tissue_window:** A CT imaging window setting that enhances the contrast of soft tissues (e.g., muscles, organs, and blood vessels), allowing for better visualization of soft tissue abnormalities such as tumors, inflammation, or injuries.

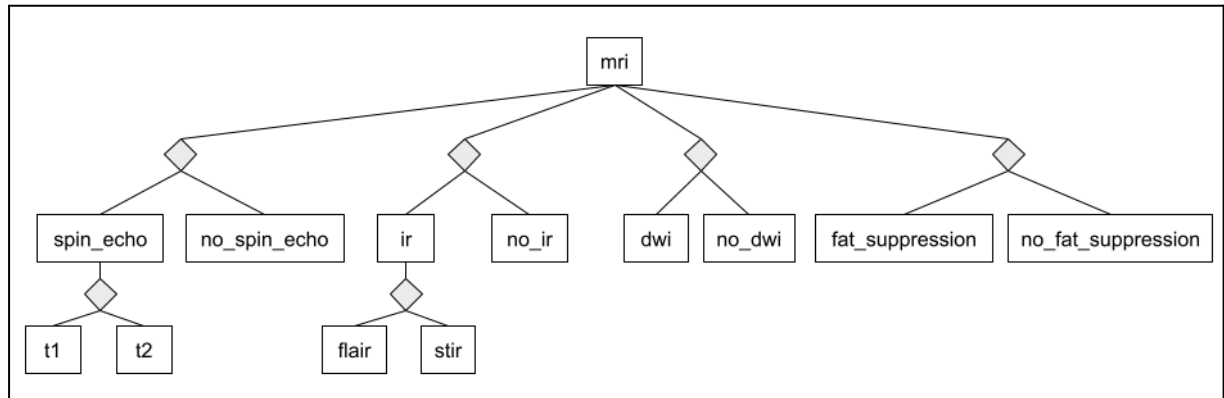
Image examples



Child classes of ct.

mri descendants

Partial class map

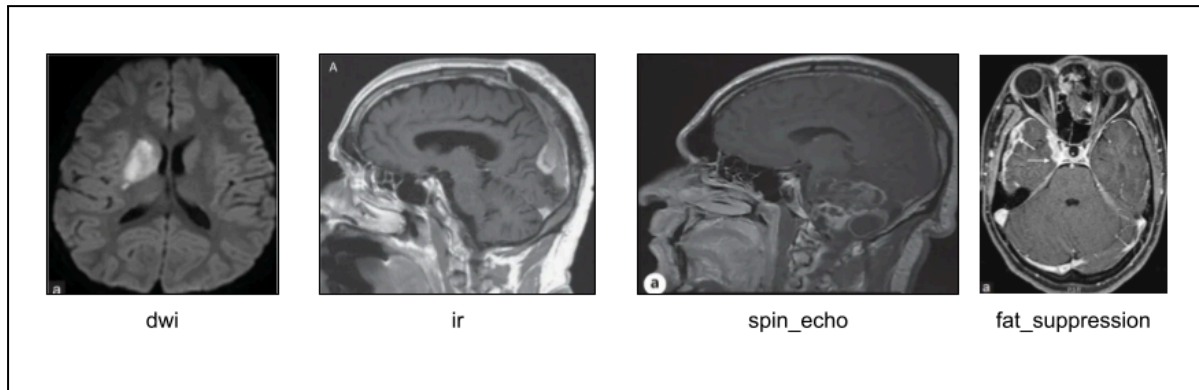


Class descriptions

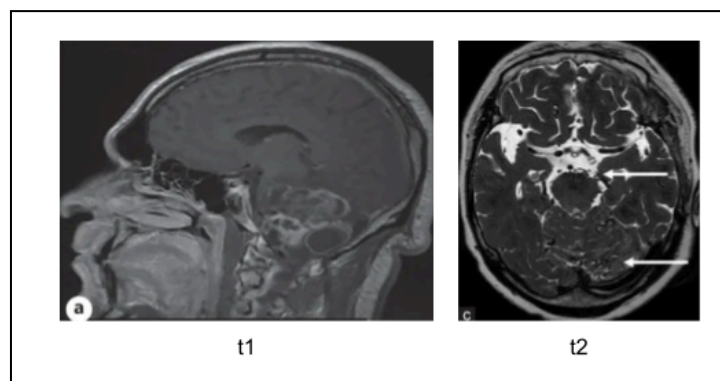
- **dwi:** Diffusion Weighted Imaging (DWI) is an MRI technique that measures the movement of water molecules in tissues. It is particularly useful for detecting acute ischemic stroke, brain tumors, and infections.
- **no_dwi:** Negative class used for MRI images that do not belong to the dwi class.
- **ir:** Inversion Recovery (IR) is an MRI sequence used to suppress specific tissue signals (such as fat) or to enhance the visibility of certain structures, like white matter lesions or tumors in the brain.
- **no_ir:** Negative class used for MRI images that do not belong to the ir class.
- **flair:** Fluid-Attenuated Inversion Recovery (FLAIR) is an MRI sequence that suppresses cerebrospinal fluid (CSF) signals, making it highly sensitive for detecting white matter lesions, multiple sclerosis, and brain tumors.
- **stir:** Short-Tau Inversion Recovery (STIR) is an MRI technique used to suppress fat signals, making it useful for imaging soft tissues and detecting muscle, tendon, or joint abnormalities, as well as for identifying certain types of tumors.
- **spin_echo:** A traditional MRI sequence that uses a 180° pulse to refocus the magnetic spins and create high-quality images with good tissue contrast. It is used in various clinical applications, including brain, musculoskeletal, and abdominal imaging.
- **no_spin_echo:** Negative class used for MRI images that do not belong to the spin_echo class.
- **t1:** T1-weighted MRI provides high-resolution images where fat appears bright and water appears dark. It is useful for evaluating anatomical structures, such as the brain, spine, and muscles.
- **t2:** T2-weighted MRI provides images where water appears bright and fat appears dark, making it particularly useful for detecting edema, inflammation, and lesions, especially in the brain and spinal cord.
- **fat_suppression:** A technique used in MRI to reduce the signal from fat tissues, thereby enhancing the visibility of other structures like tumors, lesions, or inflammatory tissues. It is often used in musculoskeletal and abdominal imaging.

- **no_fat_suppression:** Negative class used for MRI images that do not belong to the fat_suppression class.

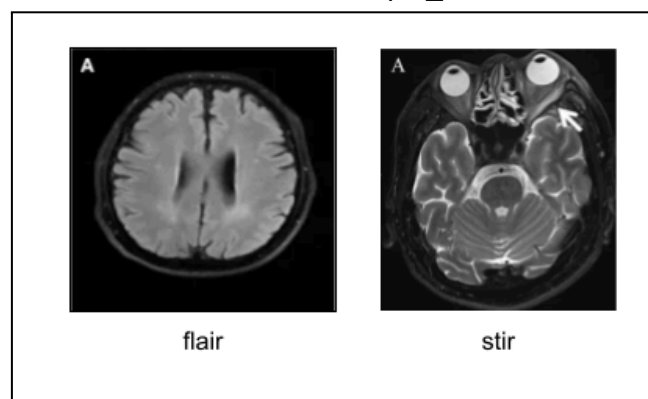
Image examples



Child classes of mri.



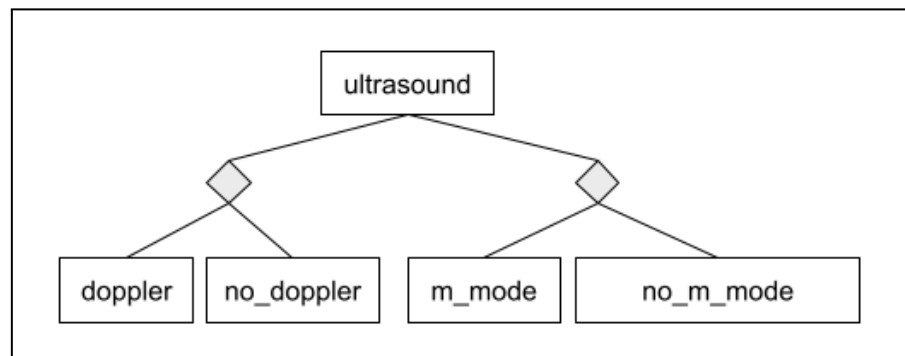
Child classes of spin_echo.



Child classes of ir.

ultrasound descendants

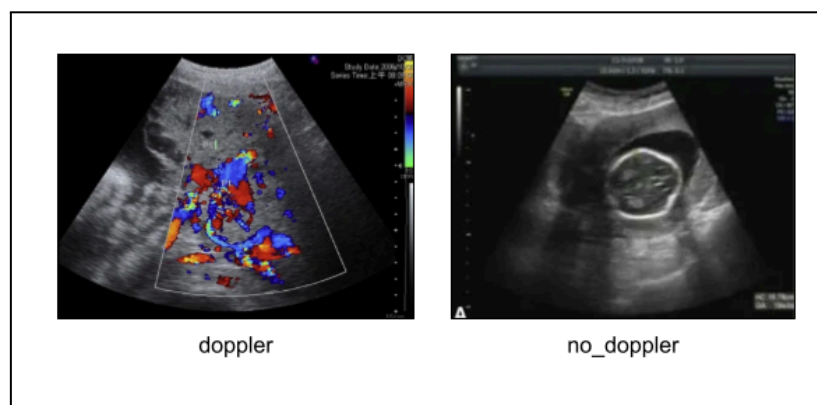
Partial class map



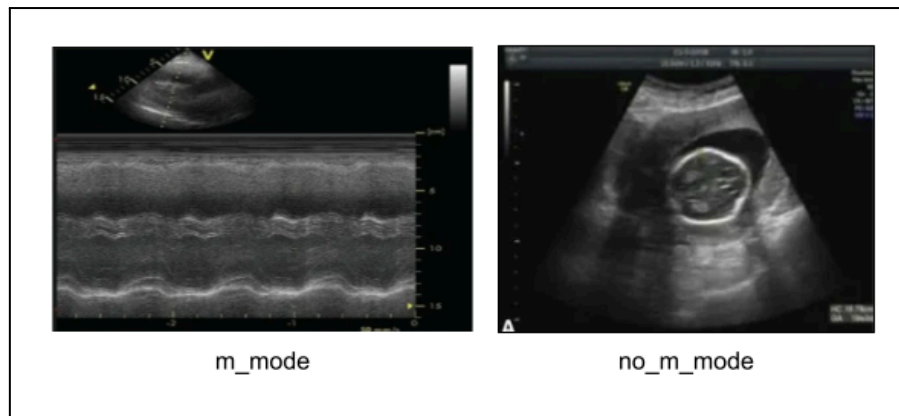
Class descriptions

- **doppler:** Doppler Ultrasound is an imaging technique used to evaluate blood flow in vessels by measuring changes in the frequency of sound waves. It is commonly used to assess the presence of blockages, clots, or abnormal blood flow in arteries and veins.
- **no_doppler:** Negative class used for ultrasound images that do not belong to the doppler class.
- **m_mode:** M-Mode Ultrasound is a technique that provides a single line of ultrasound data over time, producing a graphic representation of the movement of structures, such as heart walls or valves. It is commonly used in echocardiography to assess heart function.
- **no_m_mode:** Negative class used for ultrasound images that do not belong to the m_mode class.

Image examples



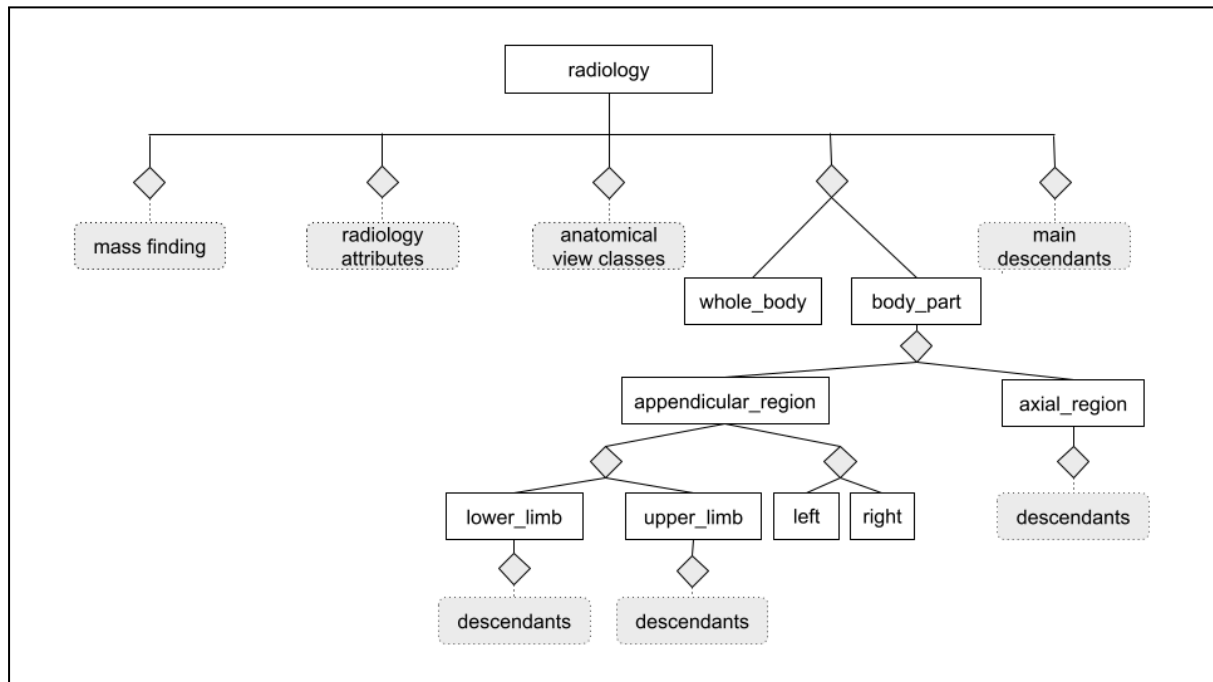
Child classes of ultrasound (doppler attribute).



Child classes of ultrasound (m_mode attribute).

anatomical_region main classes

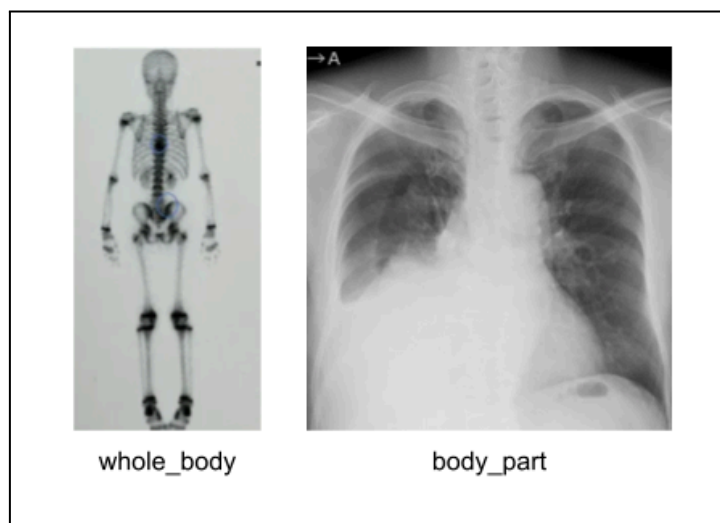
Partial class map



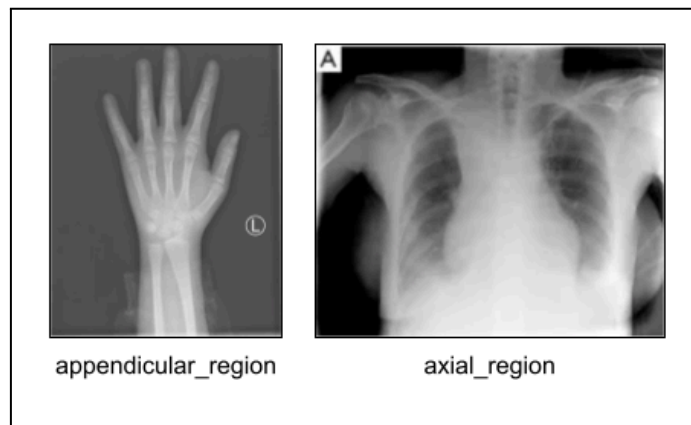
Class descriptions

- **whole_body:** Refers to the entire body, encompassing all anatomical regions, from the head to the feet, often used in imaging to represent or study the full body.
- **body_part:** Refers to a part of the body (not the whole body).
- **axial_region:** Refers to the central core of the body, including the head, neck, and trunk (torso), which houses major organs like the brain, heart, and lungs.
- **appendicular_region:** Refers to the parts of the body that include the limbs (arms and legs) and their associated structures, such as the shoulder and pelvic girdles.
- **lower_limb:** Refers to the lower extremity of the body, including the thigh, knee, leg, ankle, and foot. It encompasses the bones, muscles, and joints of the lower body.
- **upper_limb:** Refers to the upper extremity of the body, including the shoulder, arm, elbow, forearm, wrist, and hand. It is involved in movements such as lifting, gripping, and manipulating objects.
- **left:** Refers to the laterality of the left limb.
- **right:** Refers to the laterality of the right limb.

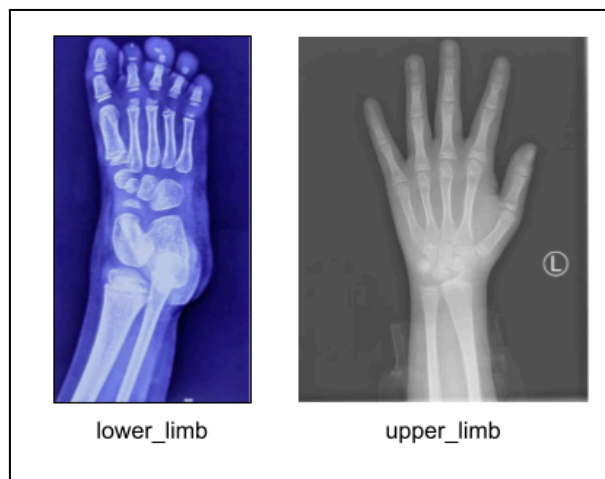
Image examples



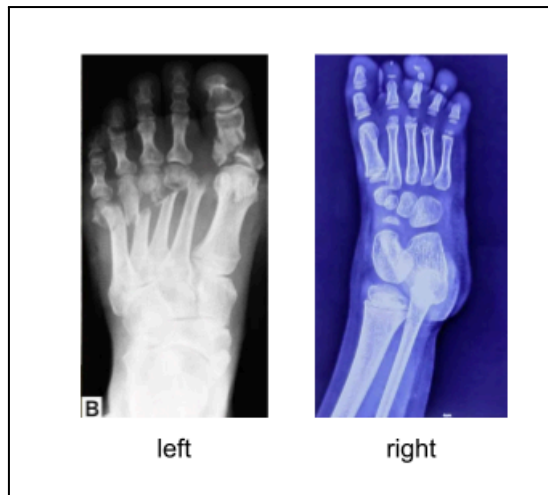
Child classes of radiology (anatomical region attribute).



Child classes of body_part.



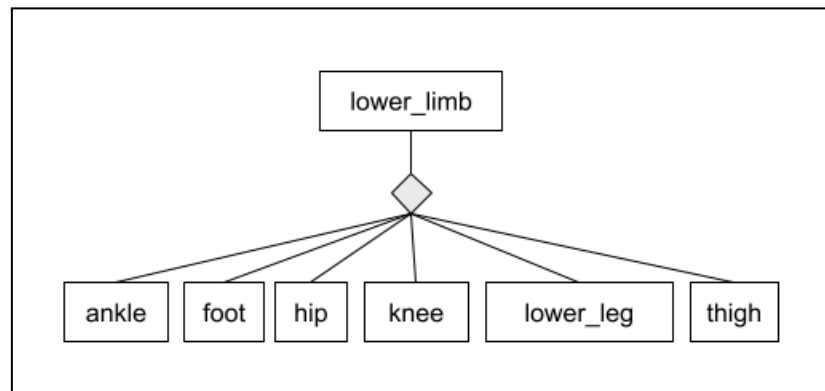
Child classes of appendicular_region (main descendants).



Child classes of appendicular_region (laterality attribute).

lower_limb descendants

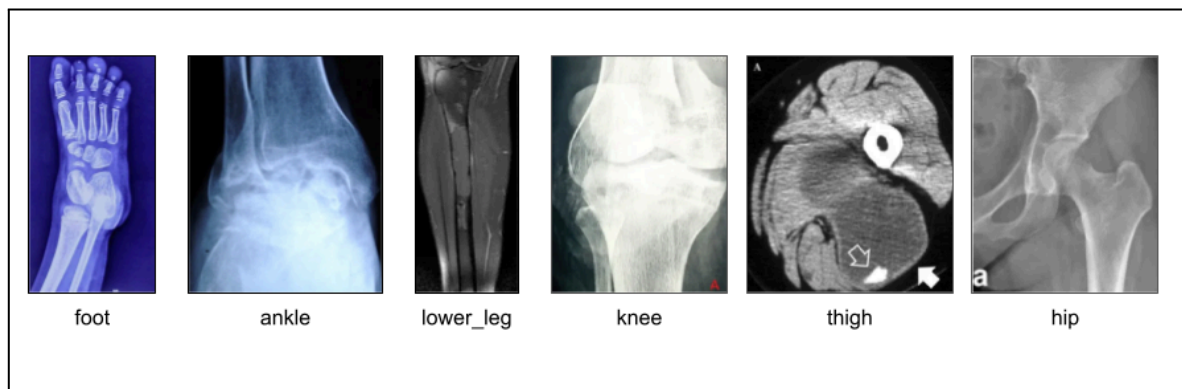
Partial class map



Class descriptions

- **foot:** The lower part of the leg that includes the toes, arch, and heel.
- **ankle:** The joint connecting the lower leg to the foot.
- **lower_leg:** Refers to the part of the leg between the knee and ankle.
- **knee:** The joint that connects the thigh to the lower leg.
- **thigh:** The upper portion of the leg, extending from the hip to the knee.
- **hip:** The joint connecting the thigh to the pelvis.

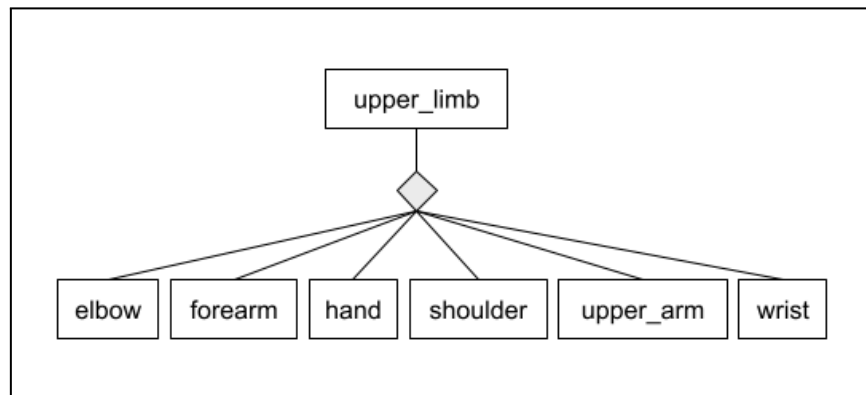
Image examples



Child classes of lower_limb.

upper_limb descendants

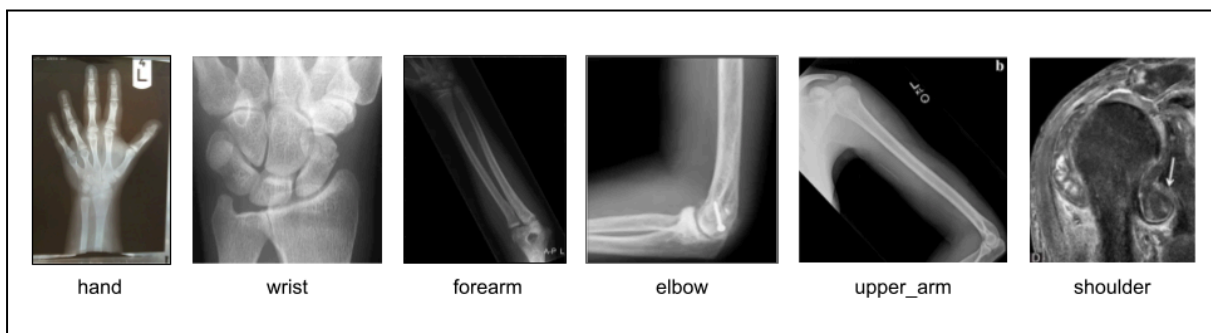
Partial class map



Class descriptions

- **hand:** The distal part of the upper limb.
- **wrist:** The joint between the forearm and hand.
- **forearm:** The portion of the arm between the elbow and wrist.
- **elbow:** The joint connecting the upper arm to the forearm.
- **upper_arm:** The section of the arm between the shoulder and the elbow.
- **shoulder:** The joint that connects the upper arm to the torso.

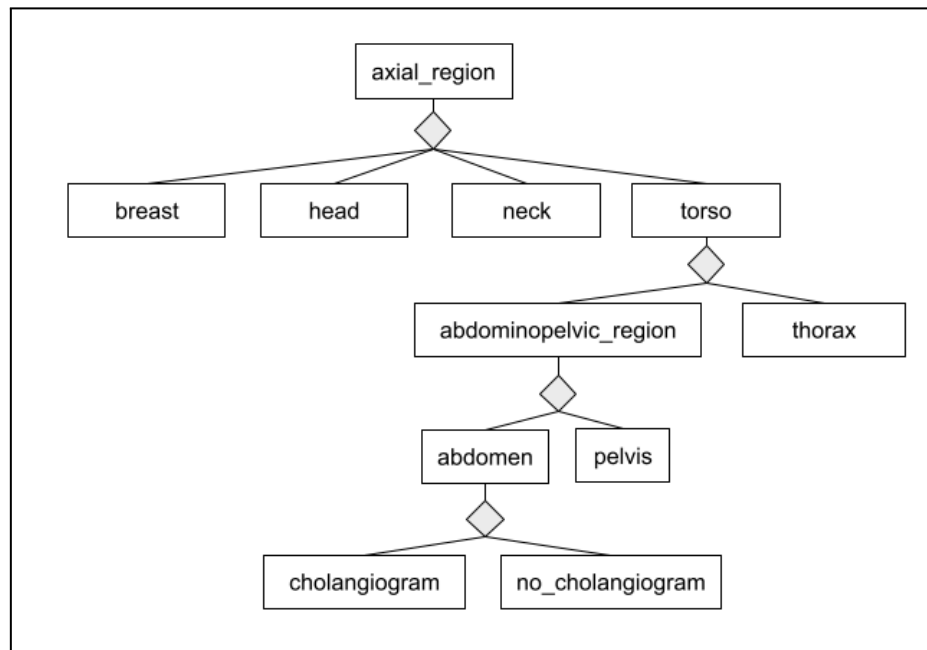
Image examples



Child classes of upper_limb.

axial_region descendants

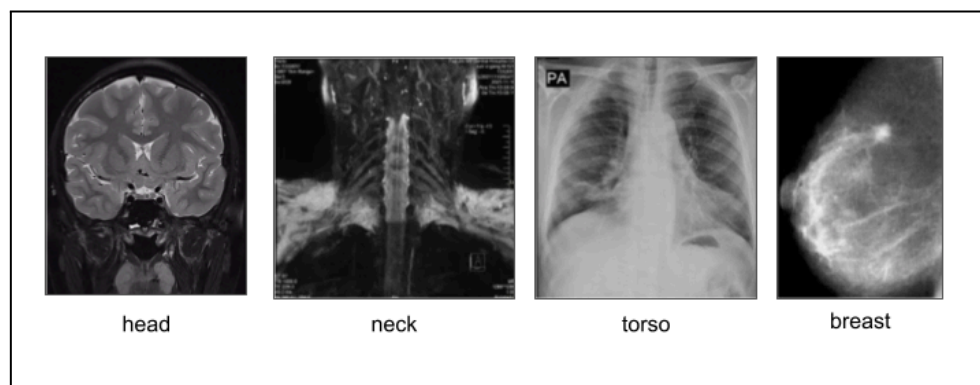
Partial class map



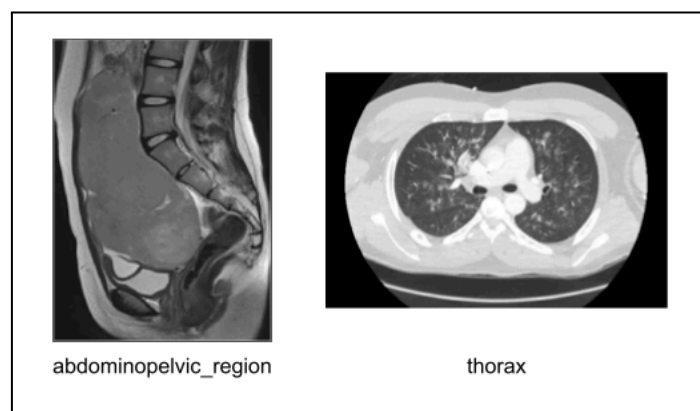
Class descriptions

- **head:** The uppermost part of the body. This class does not include images related to the eyes, as they are contained in the ophthalmic_image class.
- **neck:** The anatomical region connecting the head to the torso.
- **torso:** The central part of the axial body, including the thorax, the abdomen and the pelvis.
- **breast:** This class refers to images that include only the anatomical region of the chest that includes the mammary glands (such as mammographies).
- **thorax:** The chest region, including the ribs, lungs, heart, and major blood vessels.
- **abdominopelvic_region:** This class refers to both the abdomen and pelvis.
- **abdomen:** The section of the torso located between the chest and pelvis.
- **cholangiogram:** An imaging procedure used to visualize the bile ducts in the liver and gallbladder.
- **no_cholangiogram:** Negative class used for any abdomen image that is not specific to the bile ducts and gallbladder (i.e. images of the whole abdomen).
- **pelvis:** The lower part of the torso, located between the abdomen and thighs.

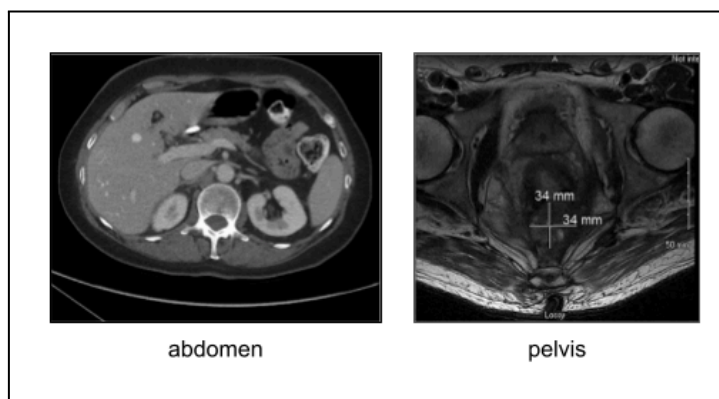
Image examples



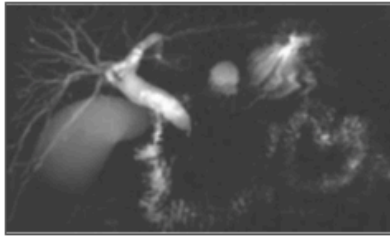
Child classes of axial_region.



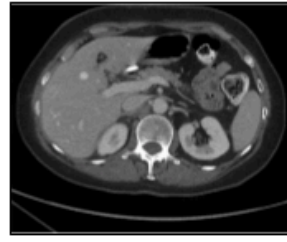
Child classes of torso.



Child classes of abdominopelvic_region.



cholangiogram

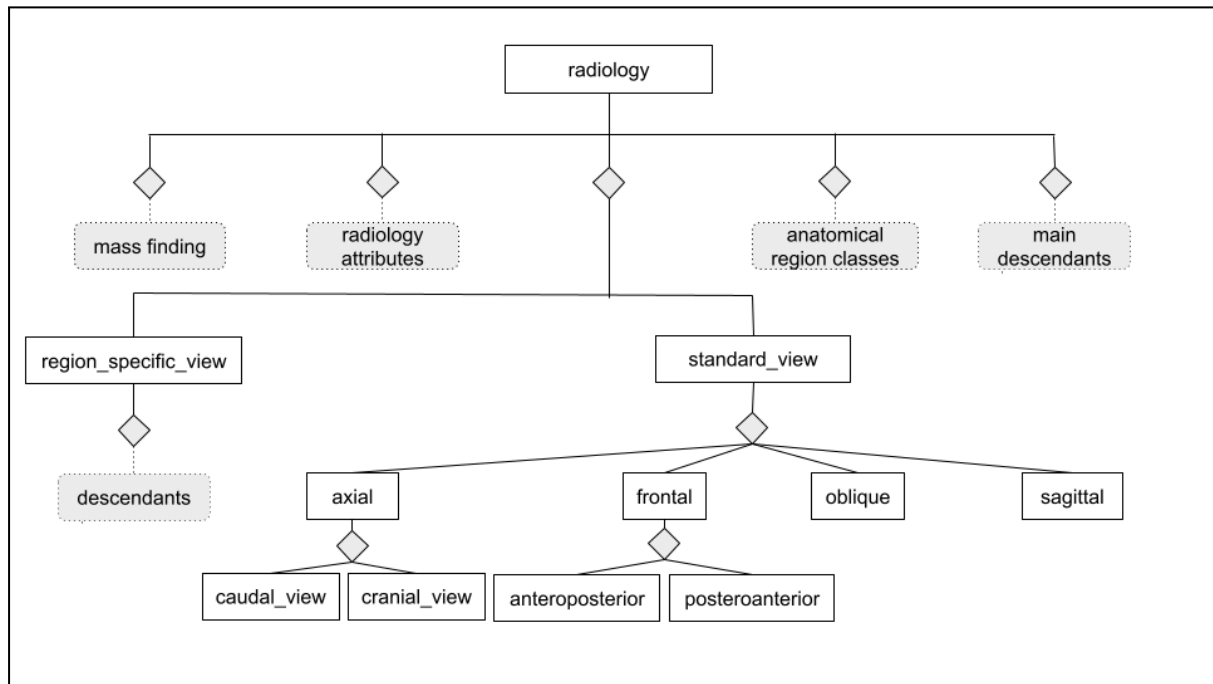


no_cholangiogram

Child classes of abdomen.

anatomical_view main classes

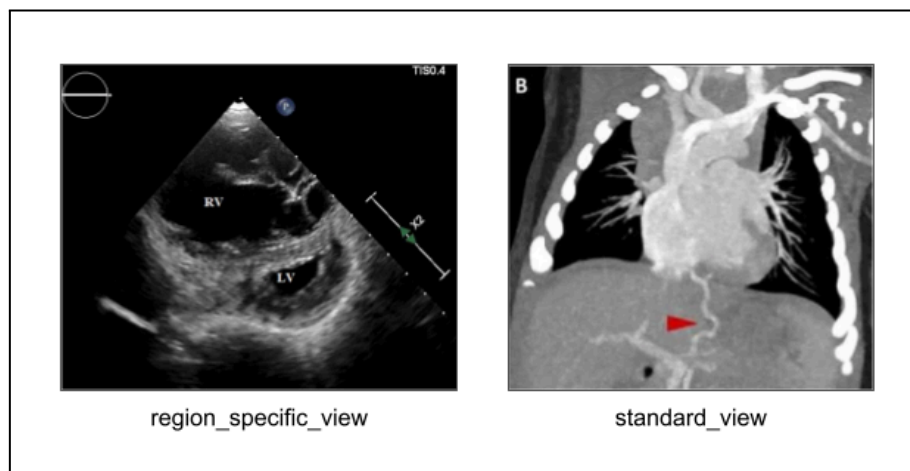
Partial class map



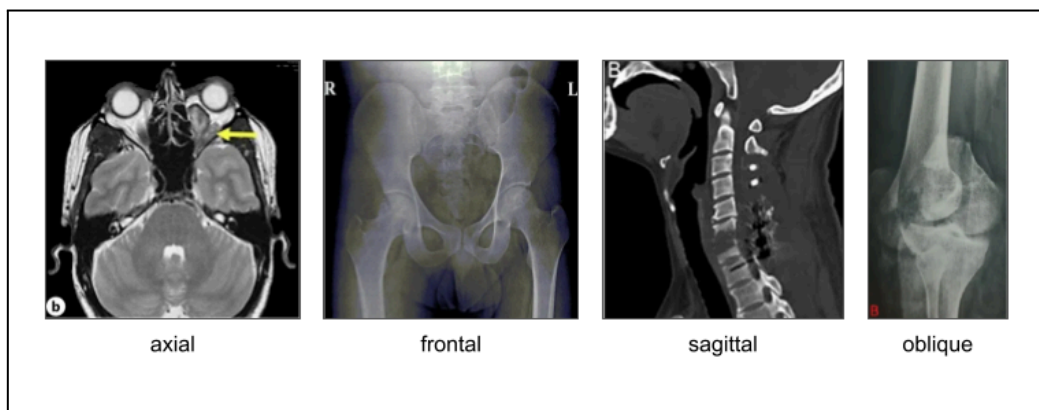
Class descriptions

- **standard_view:** View that is not specific to a given anatomical region.
- **axial:** A horizontal cross-sectional view, dividing the body into superior (top) and inferior (bottom) parts.
- **caudal_view:** A view oriented toward the lower part of the body.
- **cranial_view:** A view oriented toward the upper part of the body.
- **frontal:** A view along the frontal (coronal) plane, dividing the body into front (anterior) and back (posterior) sections.
- **anteroposterior:** An anatomical view from front (anterior) to back (posterior).
- **posteroanterior:** An anatomical view from back (posterior) to front (anterior).
- **oblique:** A slanted view, not aligned to the standard anatomical planes (e.g., axial, coronal, or sagittal).
- **sagittal:** A view along the sagittal plane, dividing the body into left and right sections.
- **region_specific_view:** A view focused on a particular anatomical region.

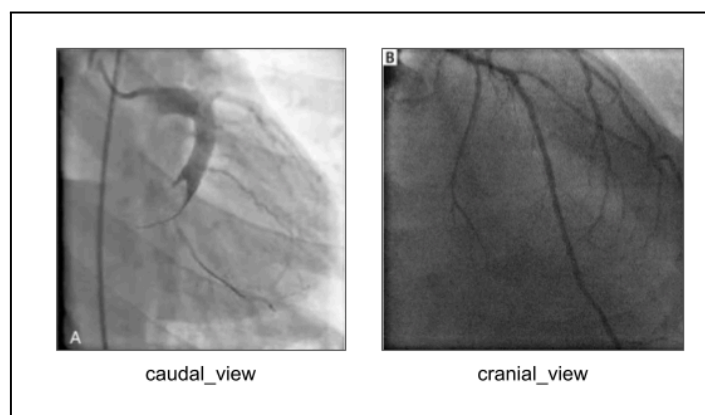
Image examples



Child classes of radiology (anatomical view attribute).



Child classes of standard_view.



Child classes of axial.



anteroposterior

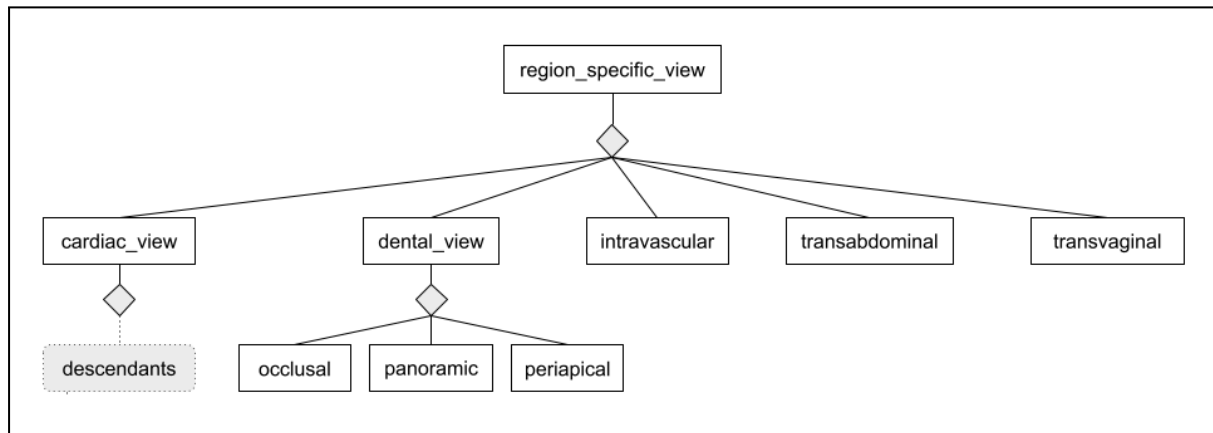


posteroanterior

Child classes of frontal.

region_specific_view descendants

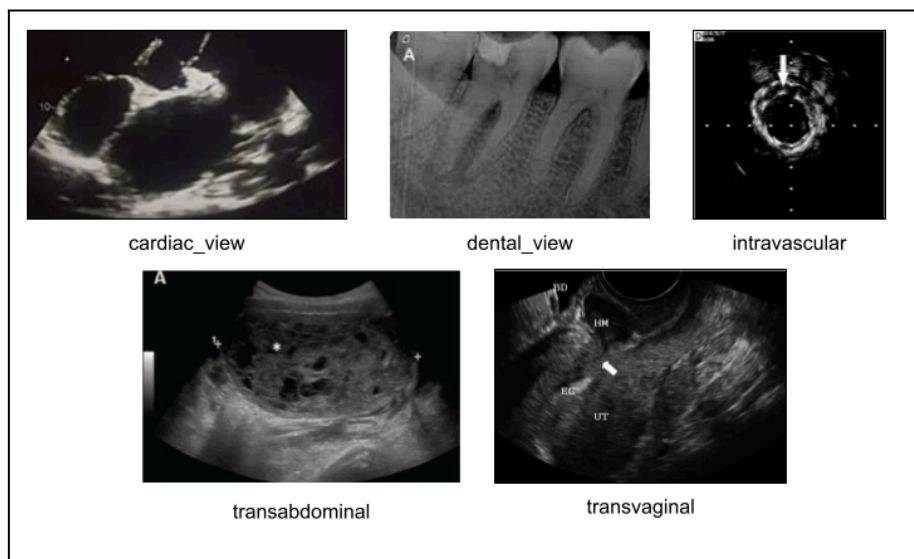
Partial class map



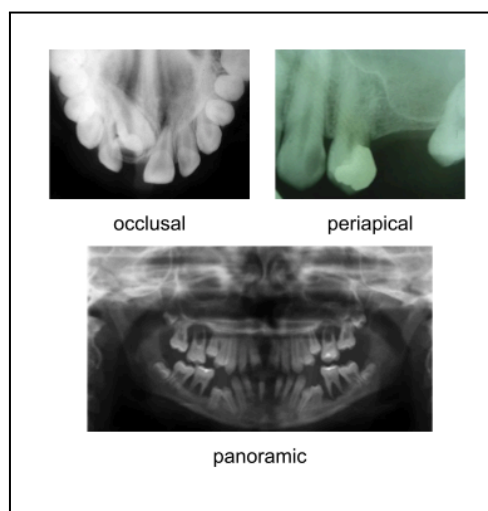
Class descriptions

- **cardiac_view:** A view specifically tailored to visualize the heart, commonly used in echocardiography or cardiac MRI.
- **dental_view:** Imaging focused on the teeth and surrounding structures.
- **occlusal:** A dental view focusing on the biting surfaces of the teeth.
- **panoramic:** A wide-angle dental X-ray that captures the entire mouth in a single image.
- **periapical:** A detailed dental X-ray focusing on the area around the tip of a tooth's root.
- **intravascular:** An imaging view from within blood vessels.
- **transabdominal:** An imaging approach performed through the abdominal wall.
- **transvaginal:** An internal imaging view using a probe inserted into the vagina.

Image examples



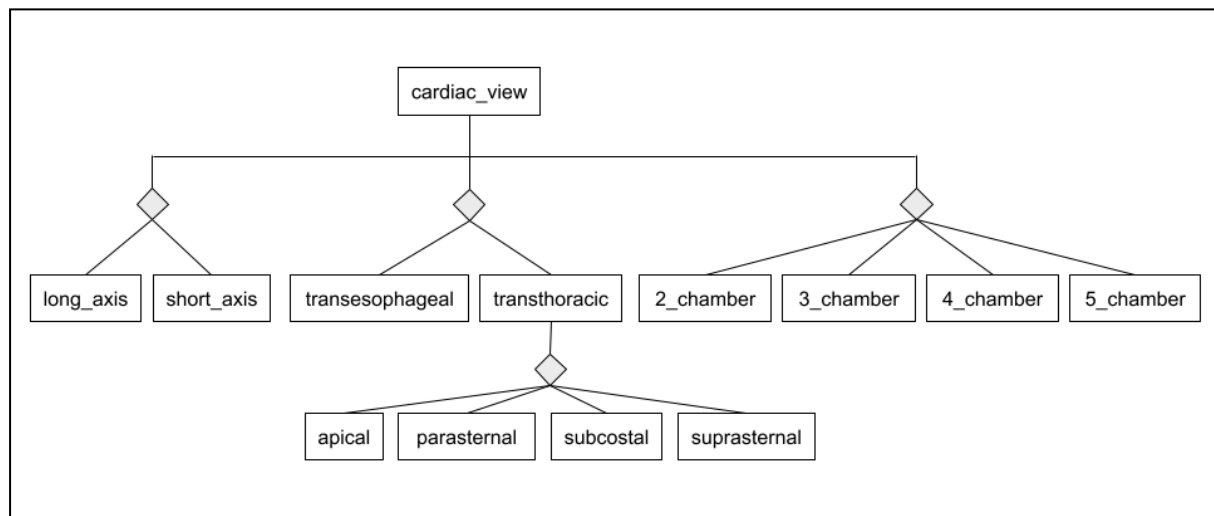
Child classes of region_specific_view.



Child classes of dental_view.

cardiac_view descendants

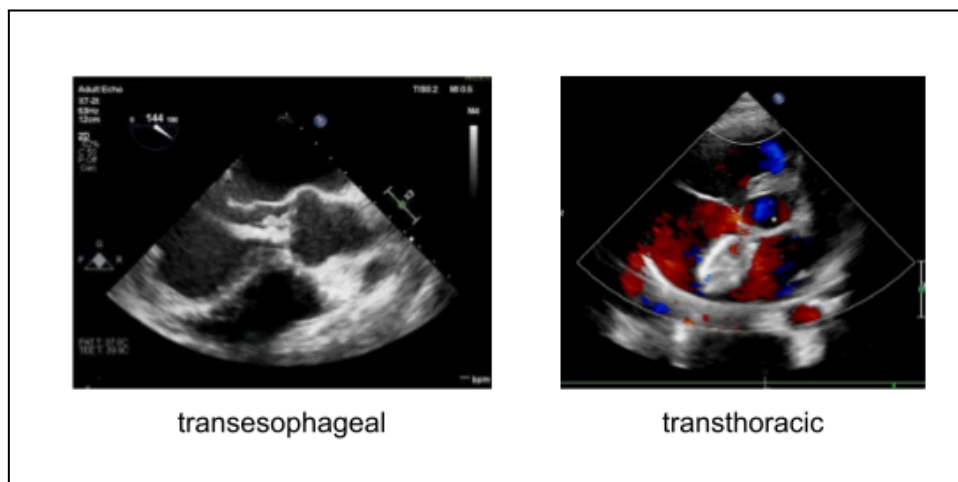
Partial class map



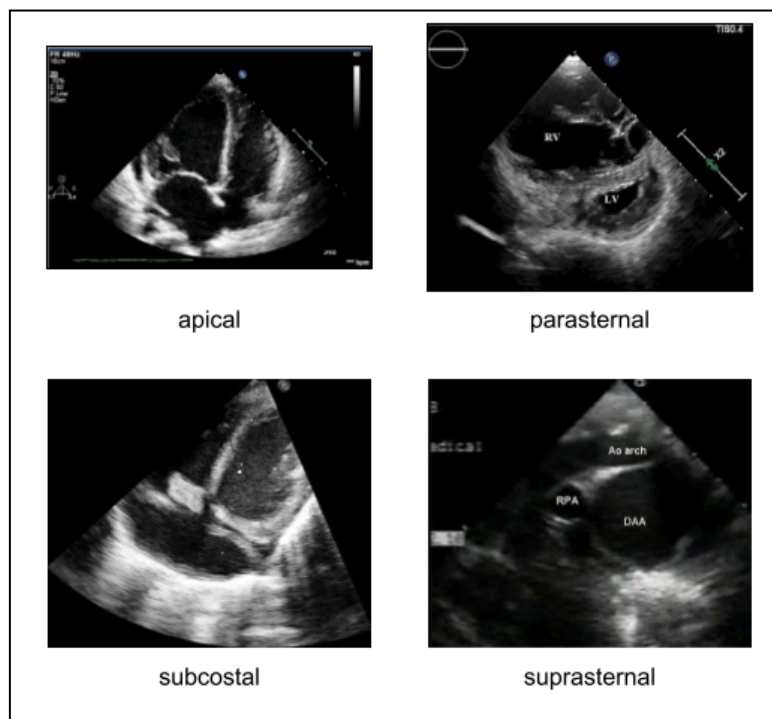
Class descriptions

- **transesophageal:** An imaging view obtained by placing an ultrasound probe in the esophagus.
- **transthoracic:** The most common echocardiographic view, where an ultrasound probe is placed on the chest wall to capture images of the heart.
- **apical:** A view obtained by placing the ultrasound probe at the apex of the heart.
- **parasternal:** A view captured by placing the probe on the chest wall beside the sternum.
- **subcostal:** An echocardiographic perspective taken from below the sternum.
- **suprasternal:** A view obtained by placing the probe at the suprasternal notch (above the sternum).
- **long_axis:** Refers to a cross-sectional view of the heart along its long axis.
- **short_axis:** Refers to a cross-sectional view of the heart perpendicular to the long axis.
- **2_chamber:** An echocardiographic view showing two heart chambers.
- **3_chamber:** This perspective visualizes three heart chambers.
- **4_chamber:** A common view displaying all four heart chambers.
- **5_chamber:** A view extending the 4-chamber perspective by including the left ventricular outflow tract (LVOT) and aortic valve.

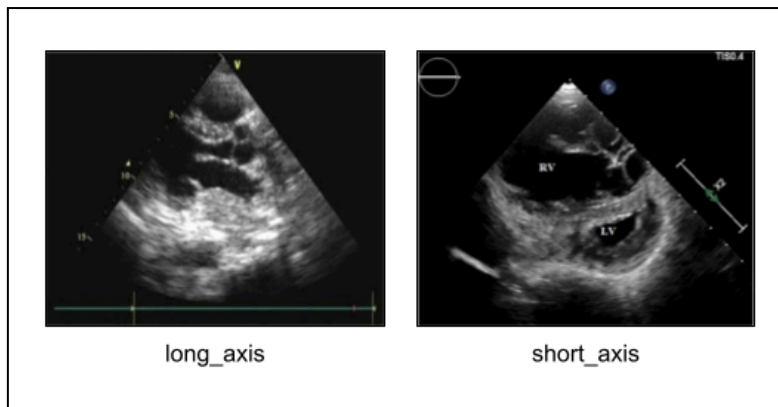
Image examples



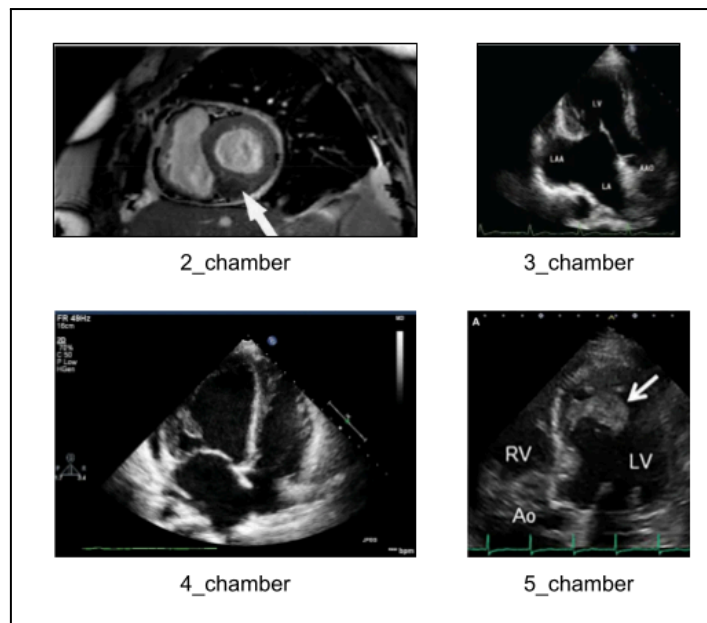
Child classes of cardiac_view (main descendants).



Child classes of transthoracic.



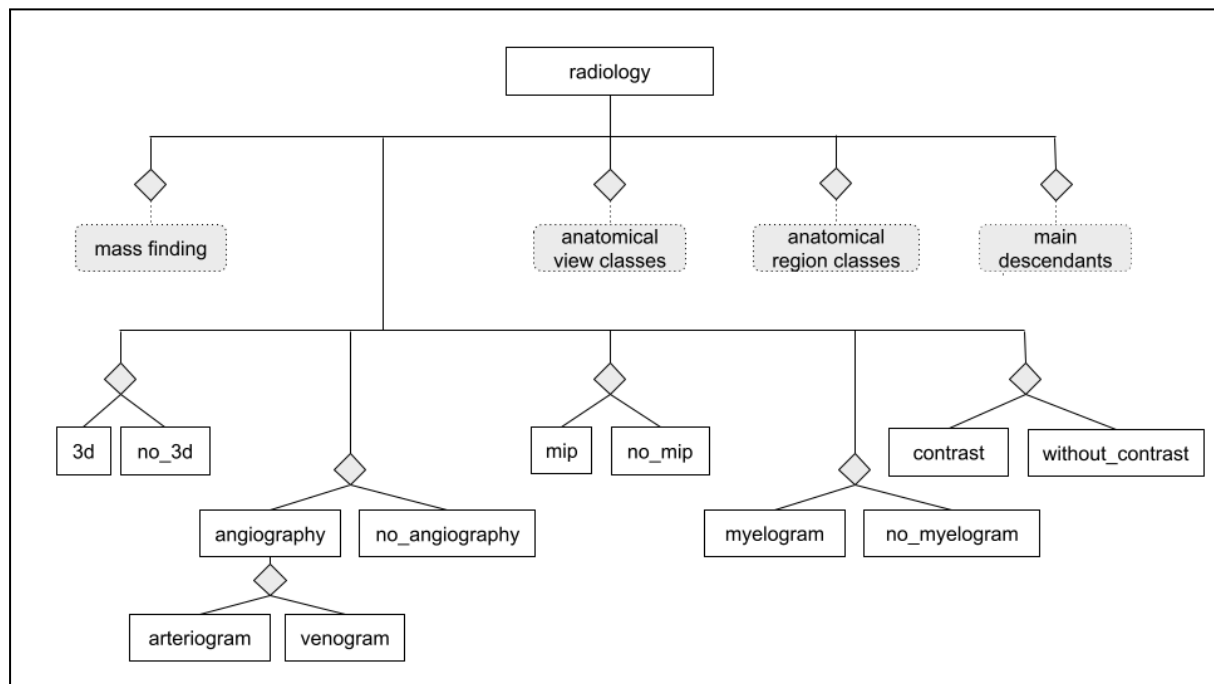
Child classes of cardiac_view (axis attribute).



Child classes of cardiac_view (chamber amount attribute).

radiology attributes

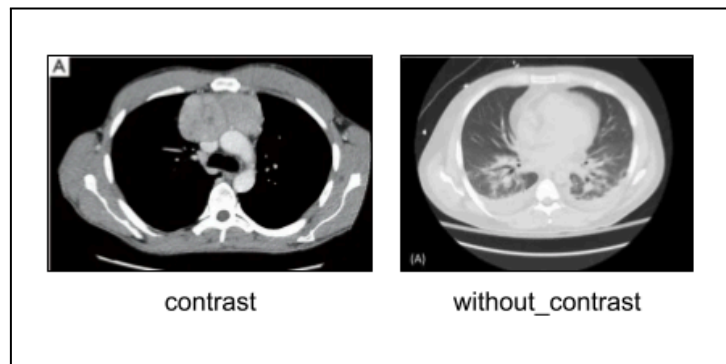
Partial class map



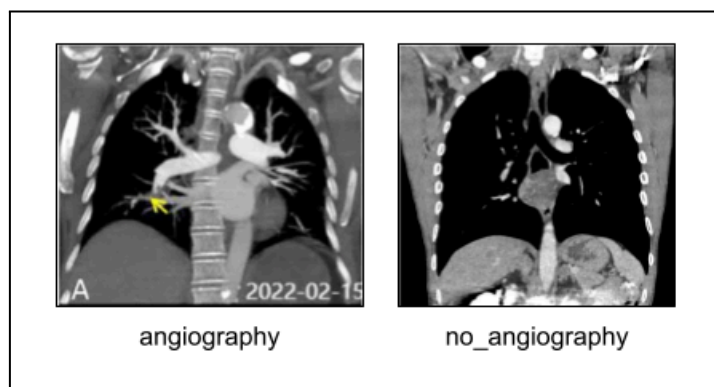
Class descriptions

- **3d**: Refers to imaging or reconstructions displayed in three dimensions, providing spatial information about anatomical structures (e.g., 3D CT or MRI models). Standard CT scans or MRI images are not considered to be 3D.
- **no_3d**: Indicates the absence of 3D rendering, typically referring to standard two-dimensional imaging.
- **angiography**: An imaging technique to visualize blood vessels.
- **arteriogram**: A specific type of angiography focused on arteries.
- **venogram**: A type of angiography that visualizes veins.
- **no_angiography**: Negative class that indicates that the imaging was performed without angiographic techniques.
- **mip**: Maximum Intensity Projection (MIP) is a post-processing technique where the brightest (highest intensity) voxels are visualized.
- **no_mip**: Negative class used for radiology images that do not belong to the mip class.
- **myelogram**: An imaging study used to visualize the spinal cord.
- **no_myelogram**: Negative class used for radiology images that do not belong to the myelogram class.
- **contrast**: Imaging performed with the use of contrast agents (e.g., iodine or gadolinium) to enhance visualization of structures or pathology.
- **without_contrast**: Imaging performed without the use of contrast agents.

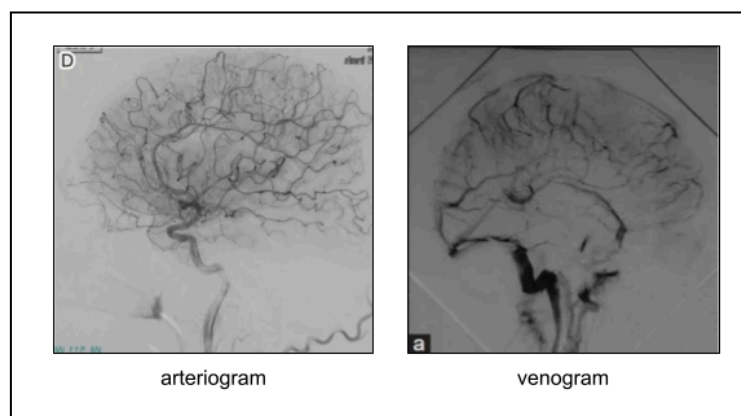
Image examples



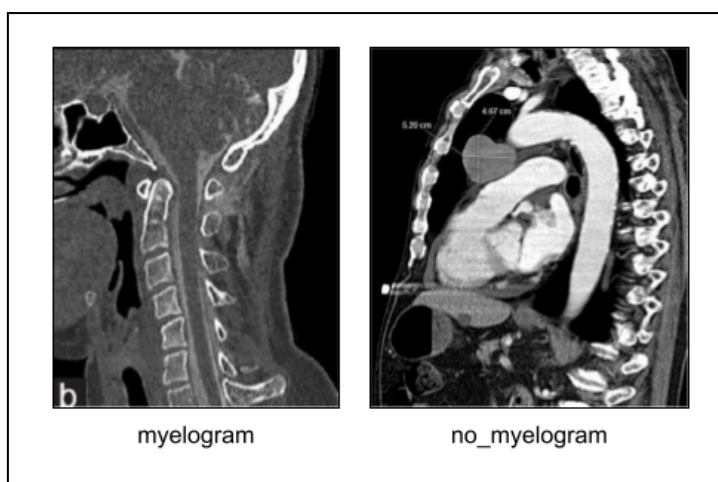
Child classes of radiology class (contrast attribute).



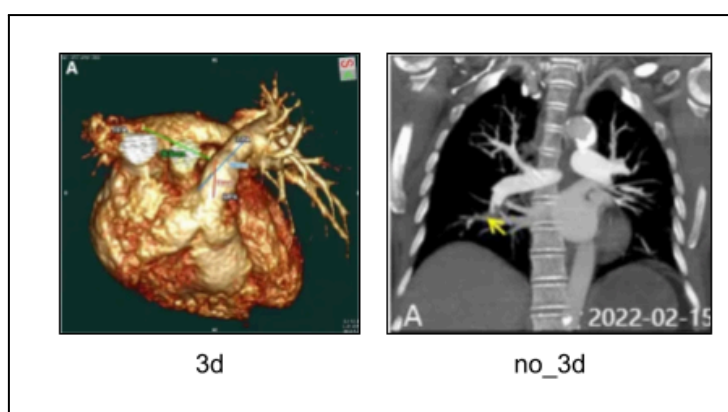
Child classes of radiology class (angiography attribute).



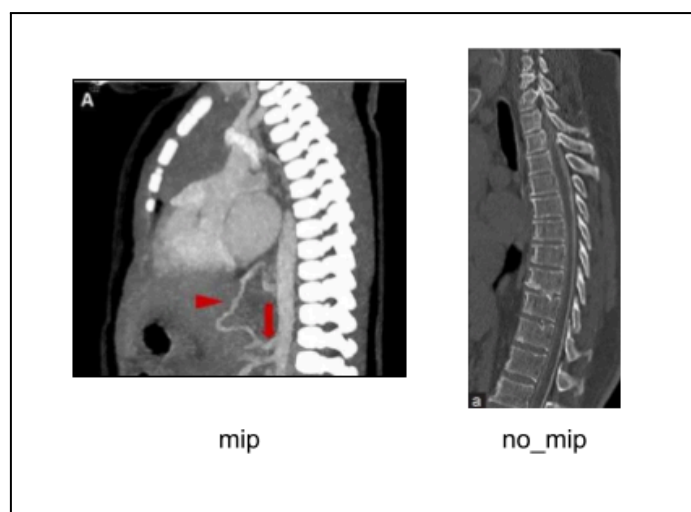
Child classes of angiography.



Child classes of radiology class (myelogram attribute).



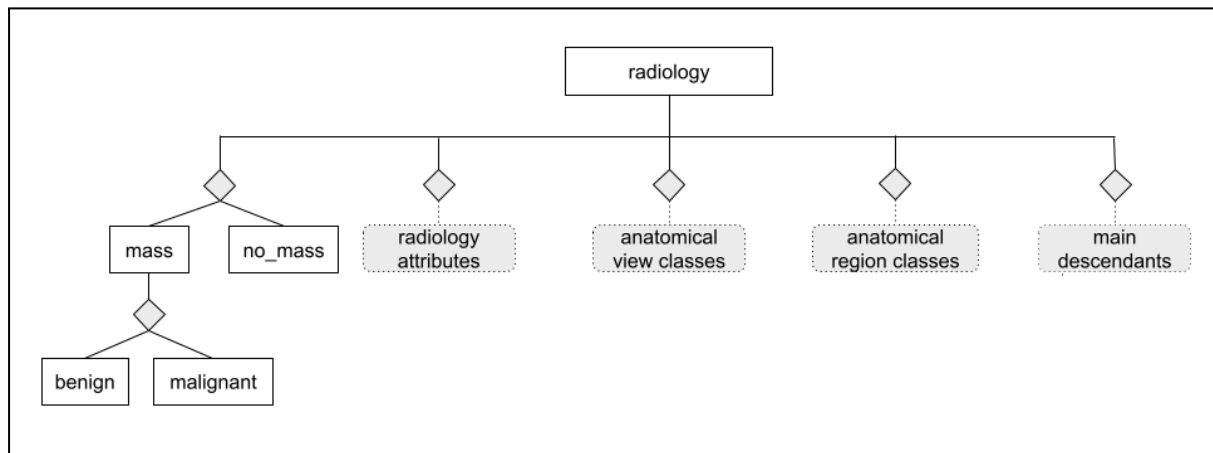
Child classes of radiology class (3d attribute).



Child classes of radiology class (mip attribute).

radiology mass_finding

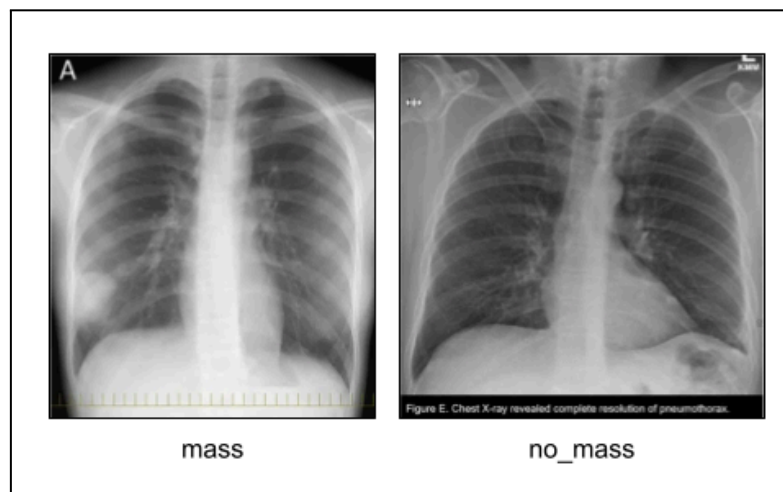
Partial class map



Class descriptions

- mass: This class is used for images that contain at least one mass finding, either benign or malignant.
- benign: Indicates that the mass is non-cancerous.
- malignant: Denotes that the mass is cancerous (either the primary tumor or metastases).
- no_mass: Negative class used for radiology images that do not contain any mass finding.

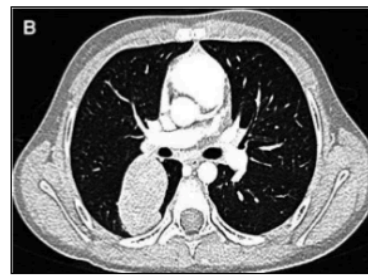
Image examples



Child classes of radiology class (mass attribute).



benign



malignant

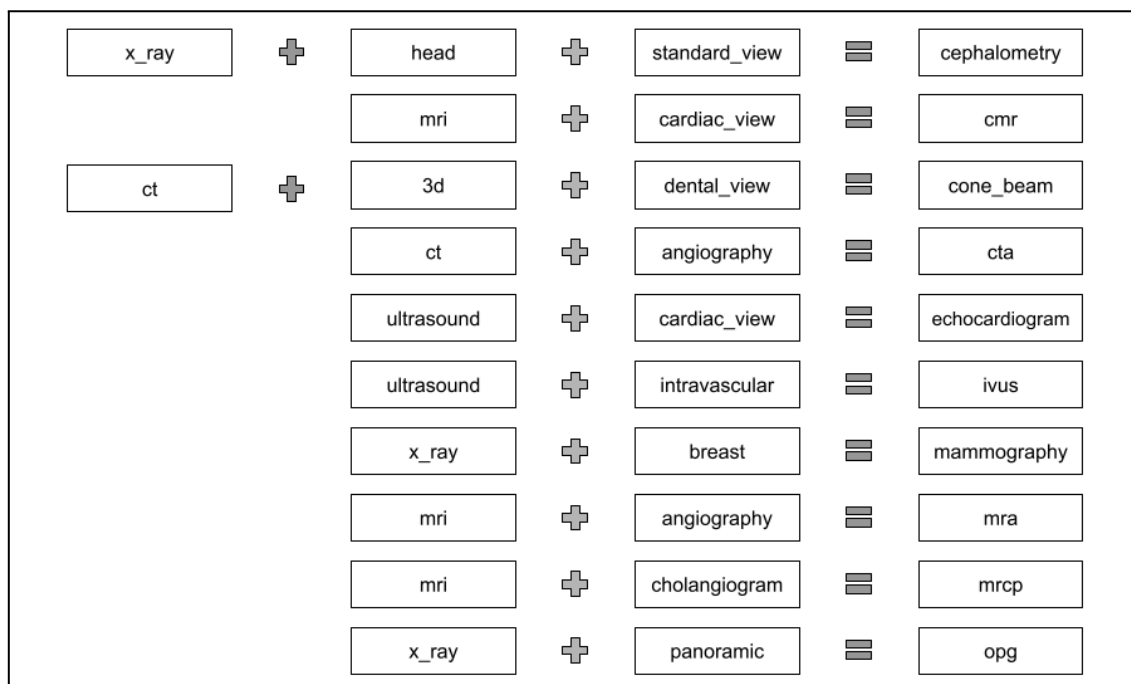
Child classes of mass.

Compound Classes

Definition

The term compound class refers to individual classes that combine multiple concepts. For example, the class echocardiography is a compound class that has two component classes: ultrasound and cardiac_view. Compound classes are not included in the class set of Multiplex classification problems; instead, they are introduced during post-processing. This approach is used in order to avoid graph structures (since component classes would have more than one parent class) and because compound classes do not involve any actual classification task: whenever an instance belongs to all component classes of a given compound class, it automatically belongs to that compound class.

Partial class map



Class descriptions

- cephalometry: Diagnostic imaging technique used to assess the skull.
- cmr: The Cardiovascular Magnetic Resonance (CMR) is a magnetic resonance imaging used to assess heart function, anatomy, and blood vessels.
- cone_beam: The Cone Beam CT is a 3D imaging technology that uses a cone-shaped X-ray beam to capture high-resolution images of the teeth, jaw, and surrounding areas.
- cta: A Computed Tomography Angiography (CTA) is an imaging technique that uses CT scans to visualize blood vessels.
- echocardiogram: An echocardiogram is an ultrasound-based imaging method used to visualize the heart's structure and function.

- ivus: An Intravascular Ultrasound (IVUS) is a medical imaging technique that uses ultrasound to view the inside of blood vessels.
- mammography: A mammography is an X-ray imaging technique used to detect breast cancer and other breast abnormalities through detailed images of breast tissue.
- mra: A Magnetic Resonance Angiography (MRA) uses magnetic resonance imaging to visualize blood vessels.
- mrcp: A Magnetic Resonance Cholangiopancreatography (MRCP) is a non-invasive MRI technique used to visualize the bile ducts, gallbladder, and pancreatic ducts.
- opg: An Orthopantomogram (OPG) is a panoramic X-ray imaging technique that provides a 2D view of the teeth, jaw, and surrounding structures.

Image examples

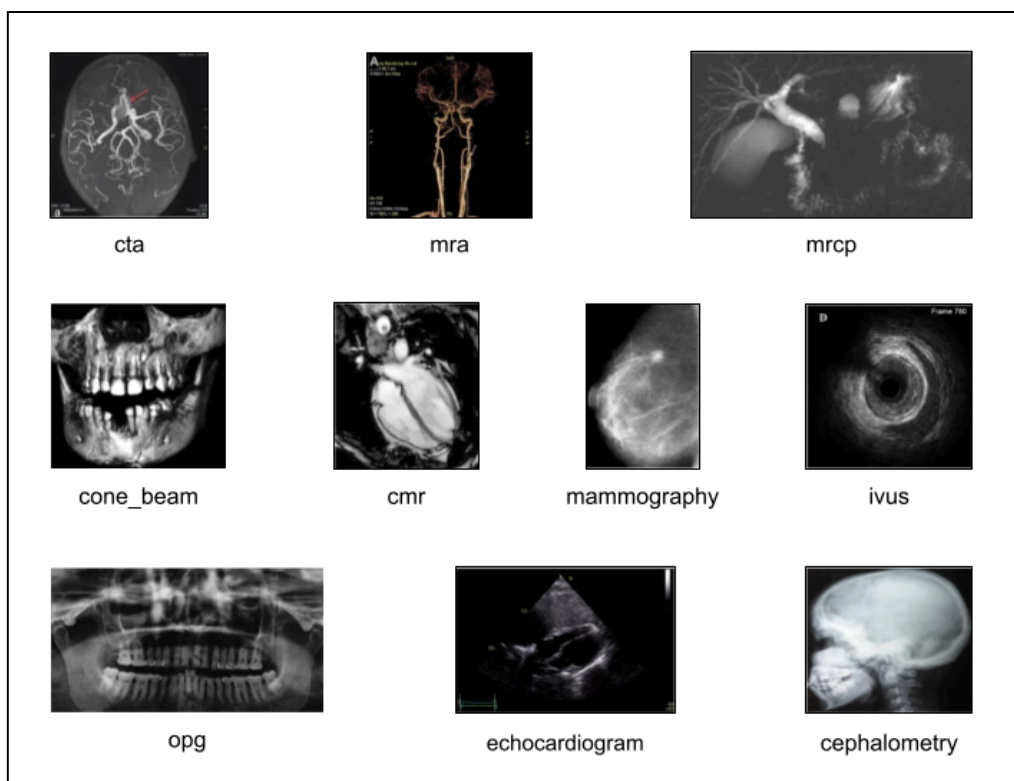


Image Credits

- 2_chamber: Solhpour, A. et al. (2014). Ventricular tachycardia from intracardiac hematoma in the setting of blunt thoracic trauma. *Indian Heart J*, 66(1):108-10. <https://doi.org/10.1016/j.ihj.2013.12.011>
- 3_chamber: Wang, B. et al. (2018). Congenital left atrial appendage aneurysm: A rare case report and literature review. *Medicine (Baltimore)*, 97(2):e9344. <https://doi.org/10.1097/MD.00000000000009344>
- 3d: Li, T. et al. (2023). Case report: Reconstruction of the long-gap unilateral absence of right pulmonary artery with contralateral pulmonary artery flap and autologous pericardial graft. *Front Cardiovasc Med*, 10:1071111. <https://doi.org/10.3389/fcvm.2023.1071111>
- 4_chamber: Hazir, K E. et al. (2023). Unexpected Course of Left Ventricular Thrombus after COVID-19 Infection in a Woman with Peripartum Cardiomyopathy Single Shield Against Double Shock. *J Saudi Heart Assoc*, 35(1):55-58. <https://doi.org/10.37616/2212-5043.1331>
- 5_chamber: Koyama, T. et al. (2020). Late-Stage Löffler's Endocarditis Mimicking Cardiac Tumor: A Case Report. *Front Cardiovasc Med*, 7:589212. <https://doi.org/10.3389/fcvm.2020.589212>
- abdomen: Portale, G. et al. (2023). Missing the main actor: Intra-operative diagnosis of agenesis of the gallbladder - When late is not 'too' late. *J Minim Access Surg*, 19(1):141-143. https://doi.org/10.4103/jmas.jmas_345_21
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- alcian_blue: Wu, S. et al. (2014). Unusual combined thymic mucoepidermoid carcinoma and thymoma: a case report and review of literature. *Diagn Pathol*, 9:8. <https://doi.org/10.1186/1746-1596-9-8>
- angiography: Safta, Y B. et al. (2019). Myxoid esophageal liposarcoma: A case report of a rare tumor. *Int J Surg Case Rep*, 60:69-71. <https://doi.org/10.1016/j.ijscr.2019.04.001>
- ankle: Sharma, S. et al. (2009). Giant cell tumor of talus: a case report. *Cases J*, 2(1):74. <https://doi.org/10.1186/1757-1626-2-74>
- anterior_segment_image: Babu, K. et al. (2010). Coin-shaped epithelial lesions following an acute attack of erythema multiforme minor with confocal microscopy findings. *Indian J Ophthalmol*, 58(1):64-6. <https://doi.org/10.4103/0301-4738.58475>

- anteroposterior: Diaz, E. et al. (2023). Thoracic Aortic Aneurysm Presenting as a Subacute Cough. *POCUS J*, 8(1):22-24. <https://doi.org/10.24908/pocus.v8i1.15894>

- apical: Hazir, K E. et al. (2023). Unexpected Course of Left Ventricular Thrombus after COVID-19 Infection in a Woman with Peripartum Cardiomyopathy Single Shield Against Double Shock. *J Saudi Heart Assoc*, 35(1):55-58. <https://doi.org/10.37616/2212-5043.1331>

- appendicular_region: Zhang, Y. et al. (2023). Case report: A case of Culler-Jones syndrome caused by a novel mutation of GLI2 gene and literature review. *Front Endocrinol (Lausanne)*, 14:1133492. <https://doi.org/10.3389/fendo.2023.1133492>

- arteriogram: Dofuku, S. et al. (2023). Successful Mechanical Thrombectomy for Isolated Internal Carotid Artery Occlusion in a Patient with Monocular Blindness: A Case Report. *NMC Case Rep J*, 10:191-195. <https://doi.org/10.2176/jns-nmc.2022-0366>

- arthroscopy: Horstmann, H. et al. (2023). All-Arthroscopic Refixation of an Osteochondral Femoral Avulsion of the Anterior Cruciate Ligament with a Laxity-Minimizing Suture. *Open Access J Sports Med*, 14:21-27. <https://doi.org/10.2147/OAJSM.S397512>

- autofluorescence: Kilicarslan, O. et al. (2023). Idiopathic Unilateral Foveal Hemorrhage in a Young Woman Without Precipitating Factors: Evaluation with Optical Coherence Tomography Angiography. *Beyoglu Eye J*, 8(2):134-138. <https://doi.org/10.14744/bej.2023.34976>

- axial: Wier, G P. et al. (2023). Orbital Venous Malformation Excision after Transcaruncular Embolization with Onyx. *Case Rep Ophthalmol*, 14(1):121-126. <https://doi.org/10.1159/000526830>

- axial_region: Hartert, M. et al. (2022). Pyopericardium and extensive mediastinal abscess following EBUS-TBNA for mediastinal staging of NSCLC: a case report. *Mediastinum*, 7:4. <https://doi.org/10.21037/med-22-13>

- b_scan: Song, W. et al. (2018). Delayed suprachoroidal hemorrhage after cataract surgery: A case report and brief review of literature. *Medicine (Baltimore)*, 97(2):e8697. <https://doi.org/10.1097/MD.00000000000008697>

- benign: Rizek, P. et al. (2011). Laparoscopic adrenalectomy for bilateral metachronous aldosteronomas. *JSLS*, 15(1):100-4. <https://doi.org/10.4293/108680811X13071180407230>

- body_part: Nakane, C. et al. (2023). A case of severe thrombocytopenia after the first exposure to rifampicin. *Respir Med Case Rep*, 42:101823. <https://doi.org/10.1016/j.rmcr.2023.101823>

- bone_window: Minghinelli, F E. et al. (2023). "Ping-pong" fracture: An exclusive entity of neonates and infants? A case analysis and literature review. *Surg Neurol Int*, 14:170. https://doi.org/10.25259/SNI_211_2023

- brain_window: Koiso, T. et al. (2022). Difficulty of diagnosing a mucor-induced aneurysm arising in segment P4 of the posterior cerebral artery - A case report. *Surg Neurol Int*, 13:111. https://doi.org/10.25259/SNI_1273_2021

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- cardiac_view: Souabni, S A. & Belhaddad, E H. (2022). Successful delivery in a woman with ventricular septal defect complicated with Eisenmenger syndrome: case report. *Pan Afr Med J*, 43:188. <https://doi.org/10.11604/pamj.2022.43.188.32982>

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- chart: Liu, S C. et al. (2023). Case Report: Anastrozole as a monotherapy for pre-pubertal children with non-classic congenital adrenal hyperplasia. *Front Endocrinol (Lausanne)*, 14:1101843. <https://doi.org/10.3389/fendo.2023.1101843>

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